# CONVOLUTIONAL NEURAL NETWORKS:

## DISCRETE CONVOLUTIONS

Convolution operations first published by D’Alembert in 1754

Discrete convolutions are matrix operations that can be used to apply **filters** to a matrix or array

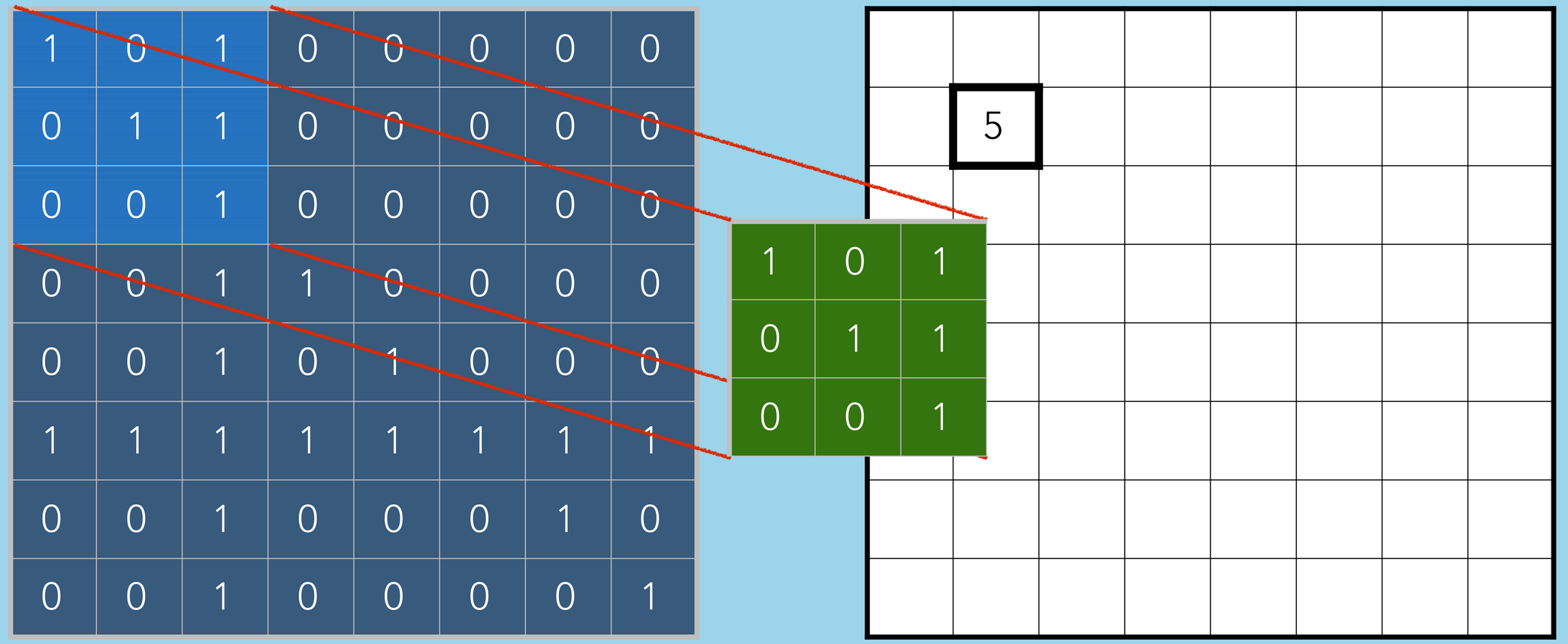
The convolutional neural network architecture was first described by Kunihiko Fukushima in 1980

# pre-defined filters

Discrete convolutions are matrix operations that can be used to apply **filters** to a matrix or array

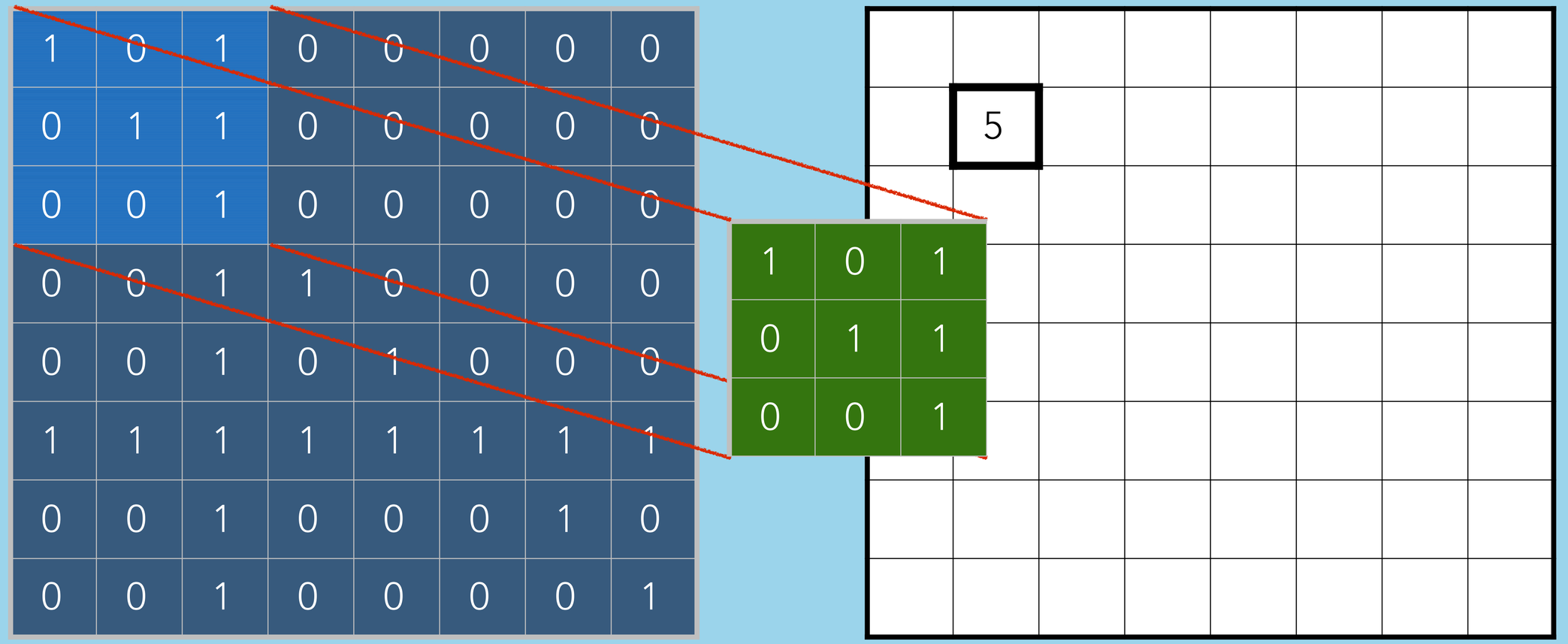
# Discrete Convolutions $C = A \circledast h$

$$C[m,n] = \sum_{j=-\omega}^{\omega} \sum_{i=-\omega}^{\omega} h[i+\omega, j+\omega] * A[m+i, n+j]$$



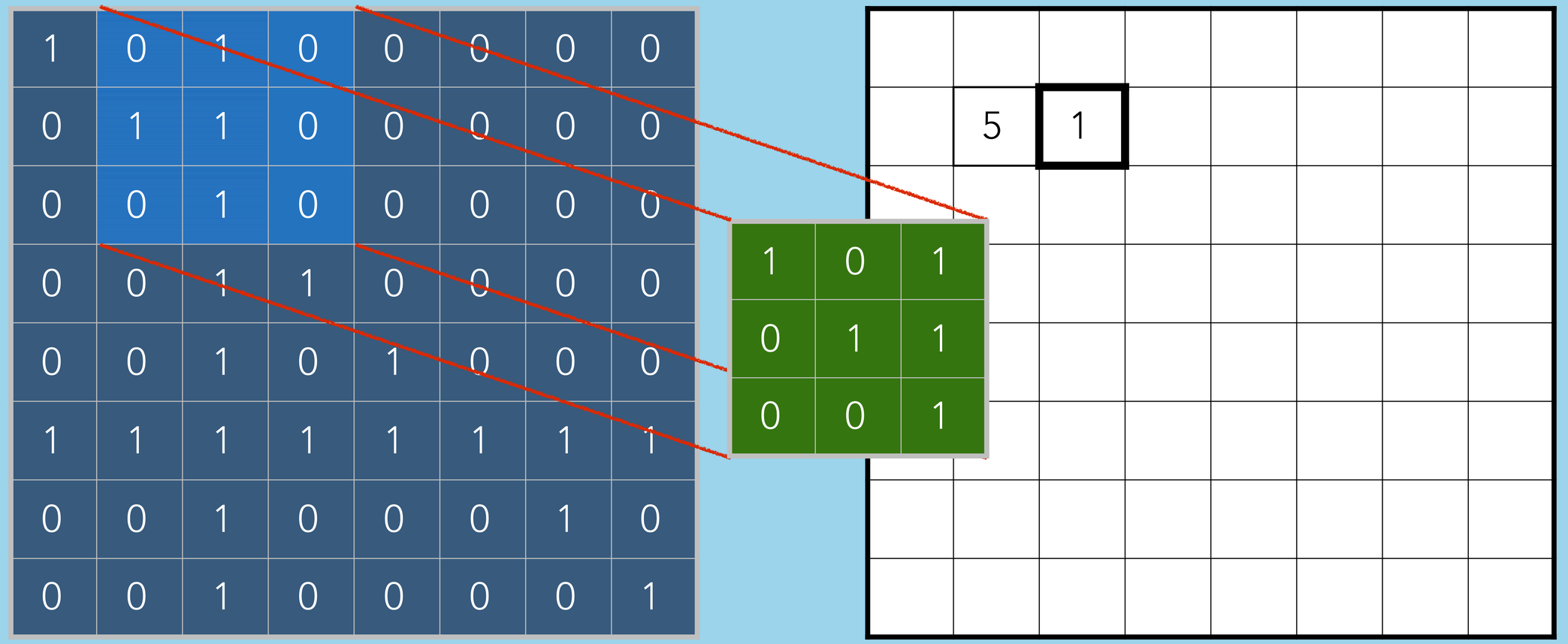
# Discrete Convolutions $C = A \circledast h$

Filter Stride Edges



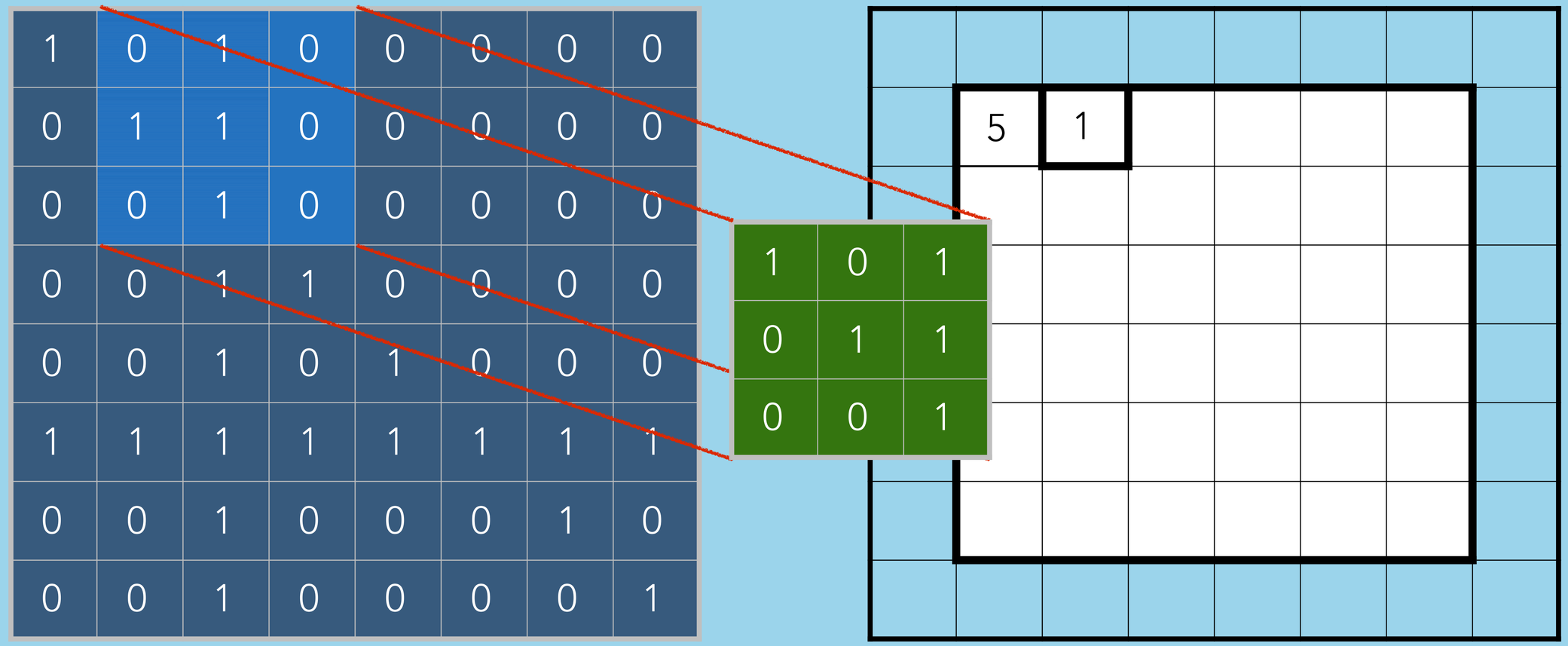
# Discrete Convolutions $C = A \circledast h$

Filter Stride Edges



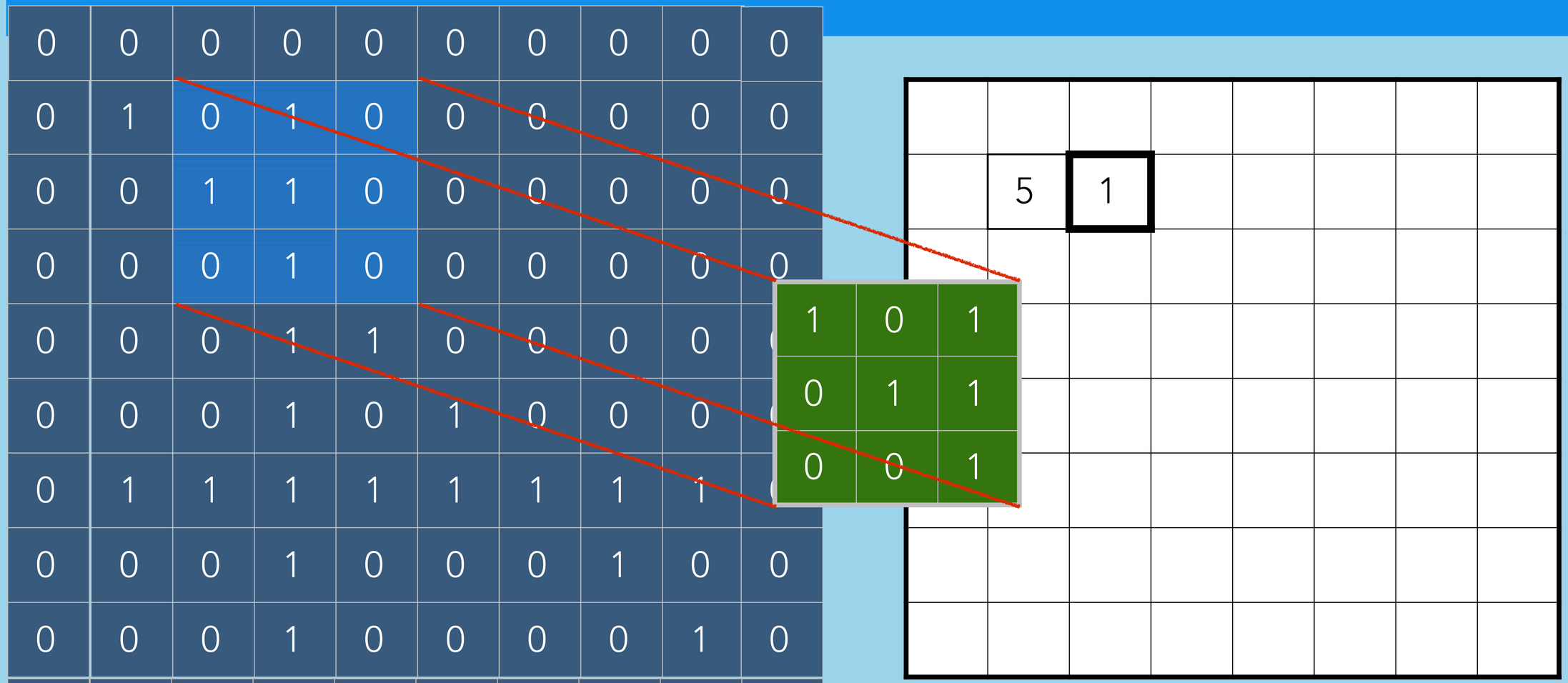
# Discrete Convolutions $C = A \circledast h$

Filter Stride Edges



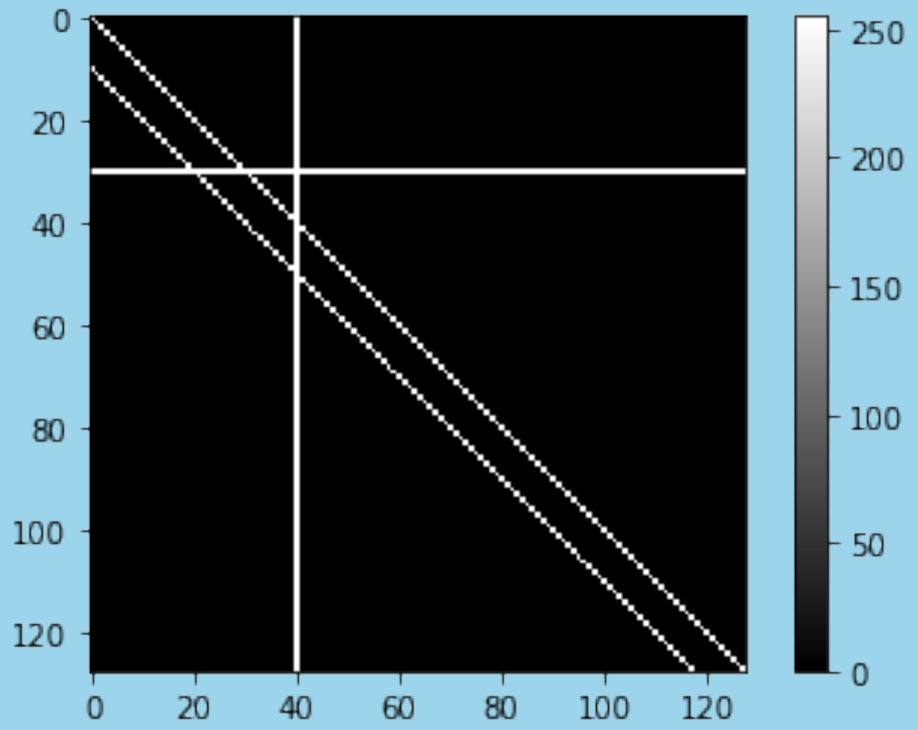
# Discrete Convolutions $C = A \circledast h$

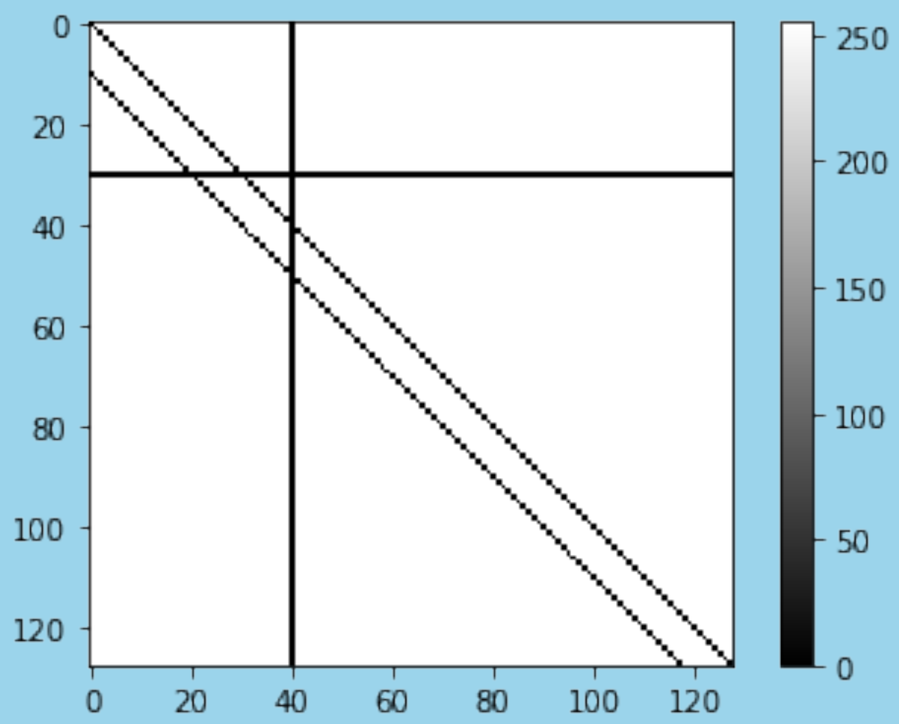
Filter Stride Edges



# Discrete Convolutions $C = A \circledast h$

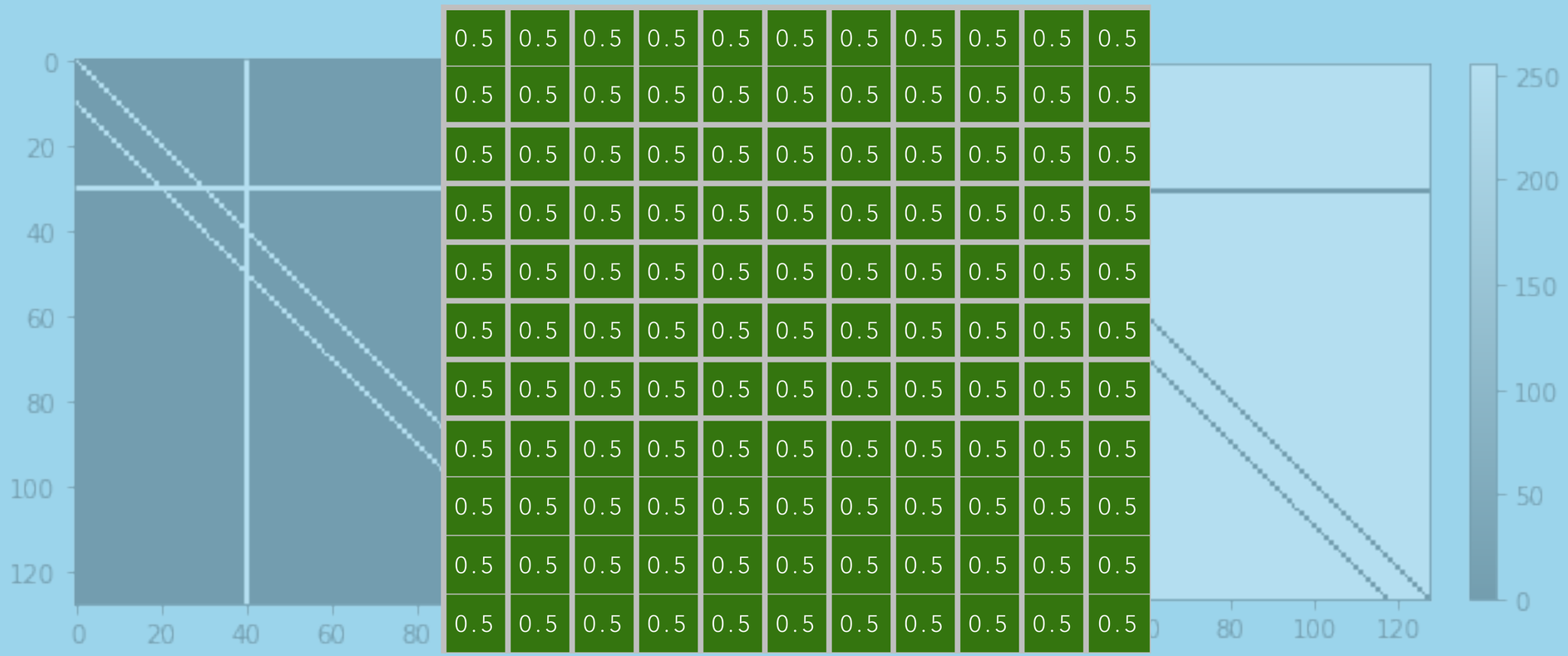
| Filter | Stride | Edges |
| --- | --- | --- |
| (11x11) | 1 | None |



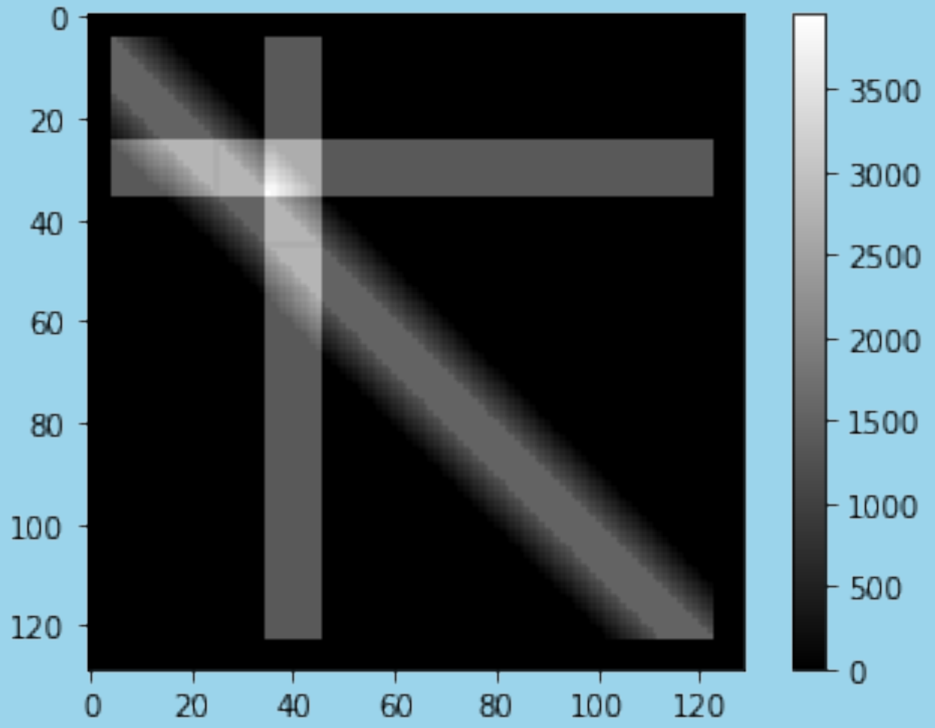
# Discrete Convolutions $C = A \circledast h$

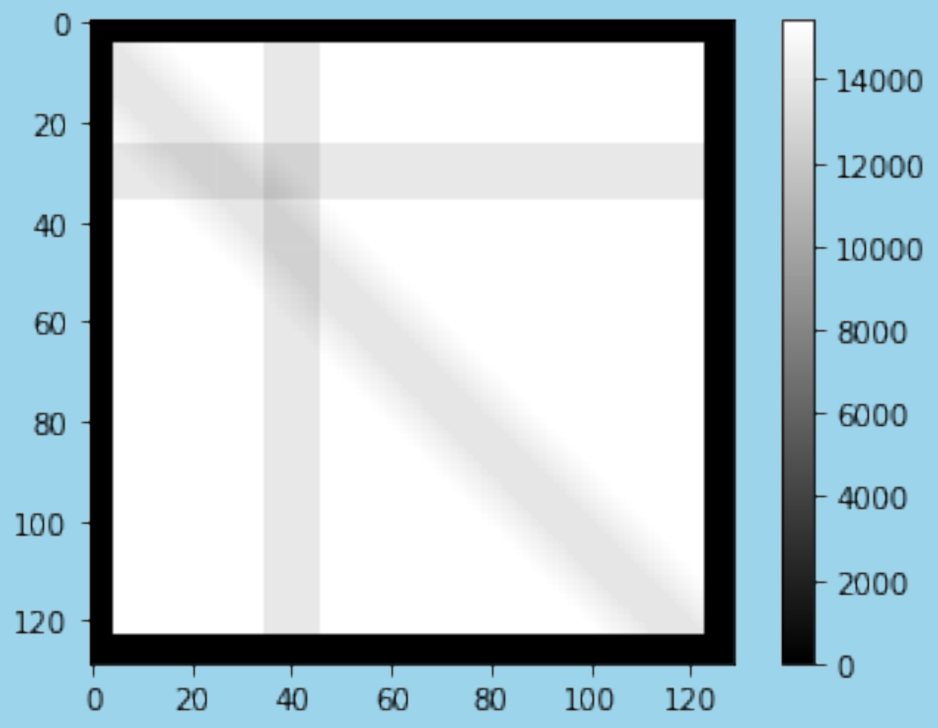
| Filter | Stride | Edges |
| --- | --- | --- |
| (11x11) | 1 | None |



# Discrete Convolutions $C = A \circledast h$

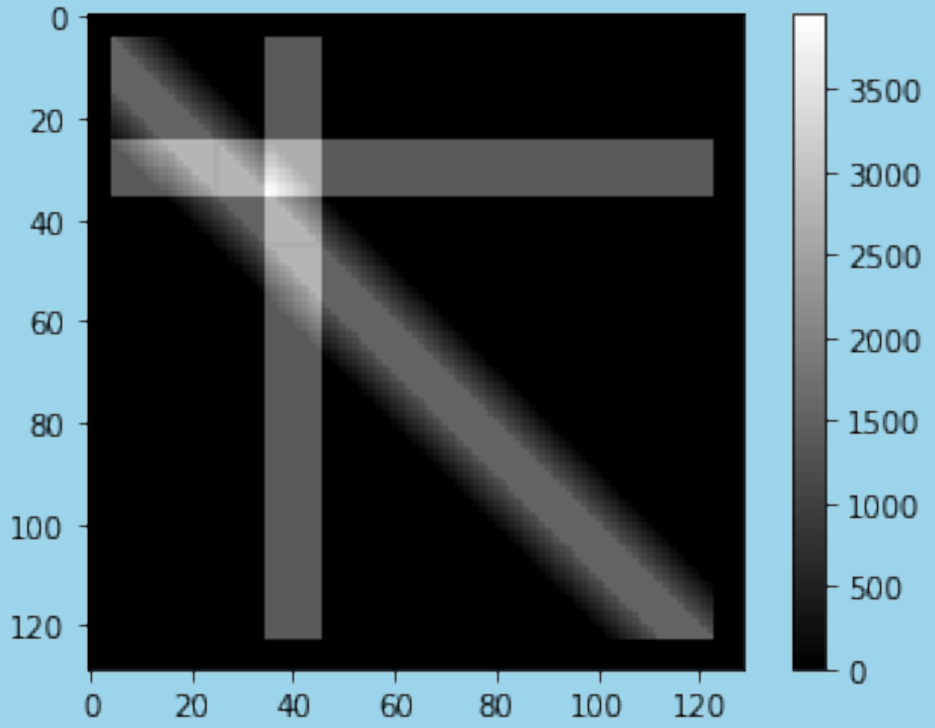
| Filter | Stride | Edges |
|---|---|---|
| (11x11) | 1 | None |

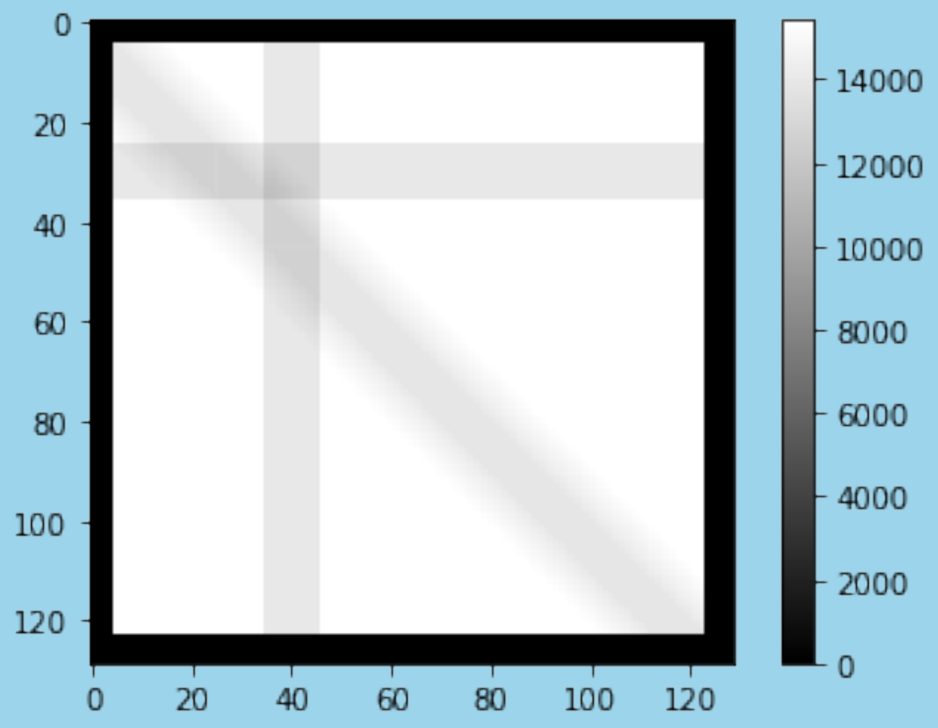


# Discrete Convolutions $C = A \circledast h$

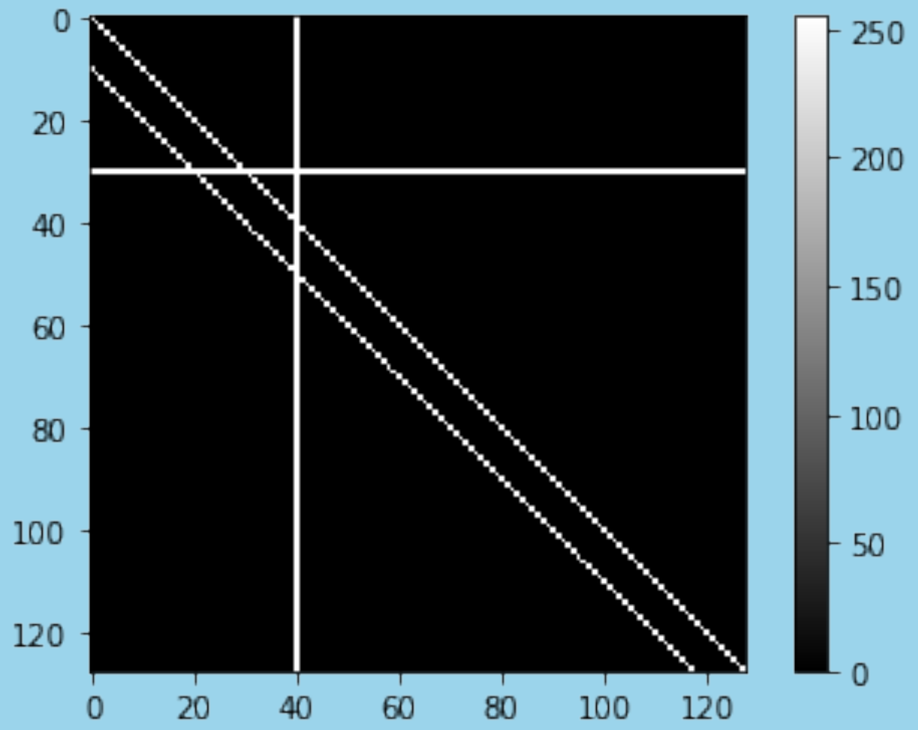
## Blurring Filter

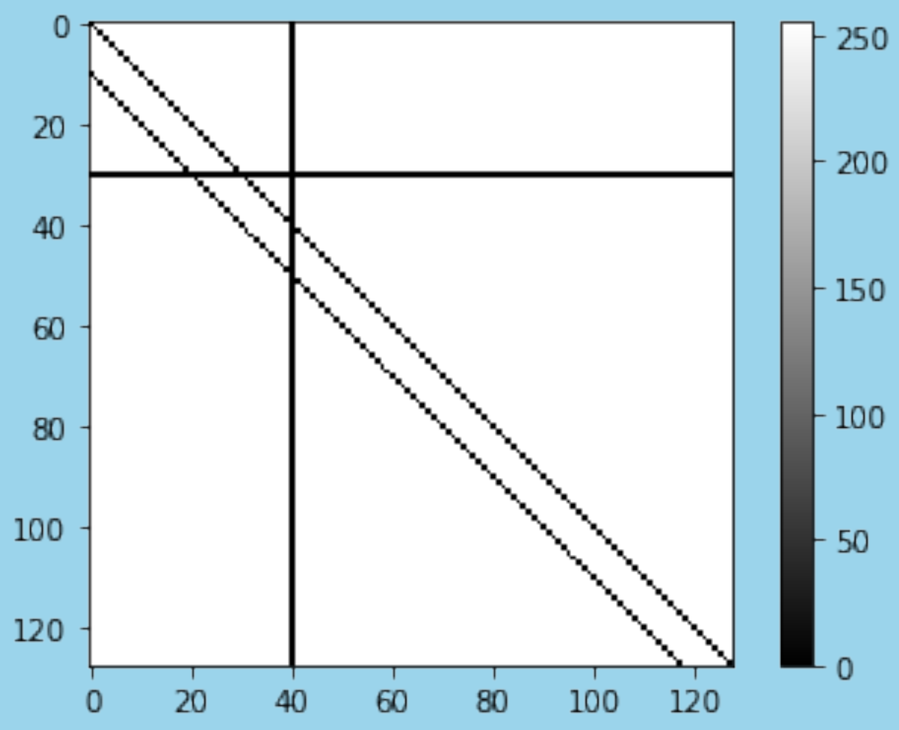


# Discrete Convolutions $C = A \circledast h$

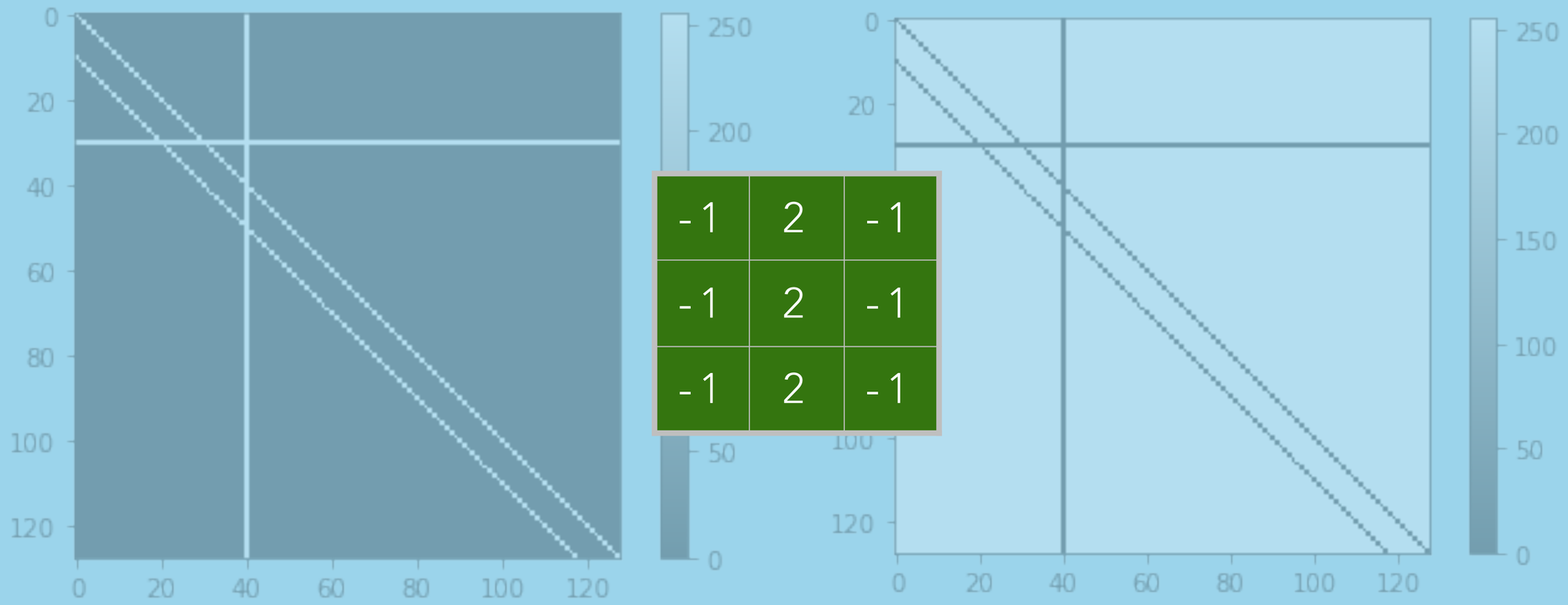
| Filter | Stride | Edges |
| --- | --- | --- |
| (3x3) | 1 | None |



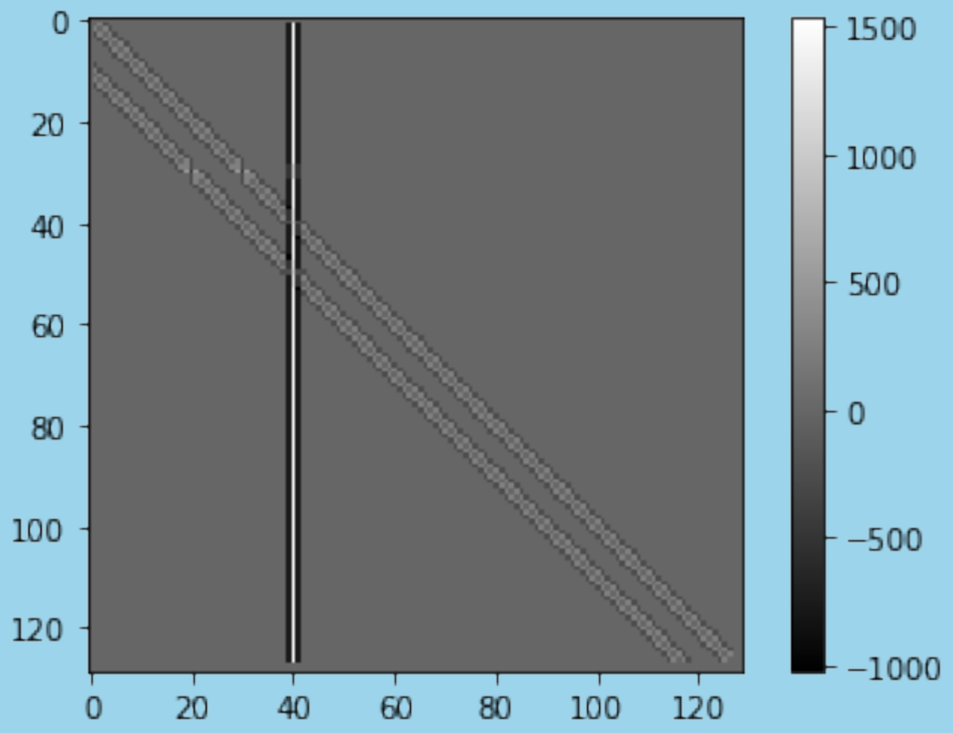
# Discrete Convolutions $C = A \circledast h$

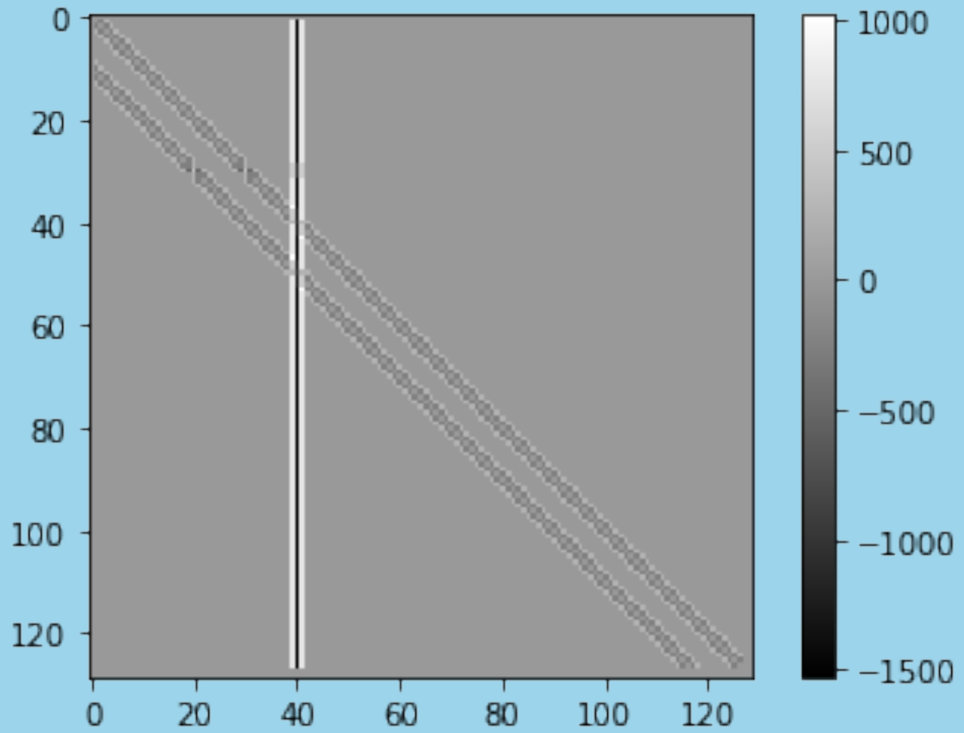
| Filter | Stride | Edges |
|---|---|---|
| (3x3) | 1 | None |



# Discrete Convolutions $C = A \circledast h$

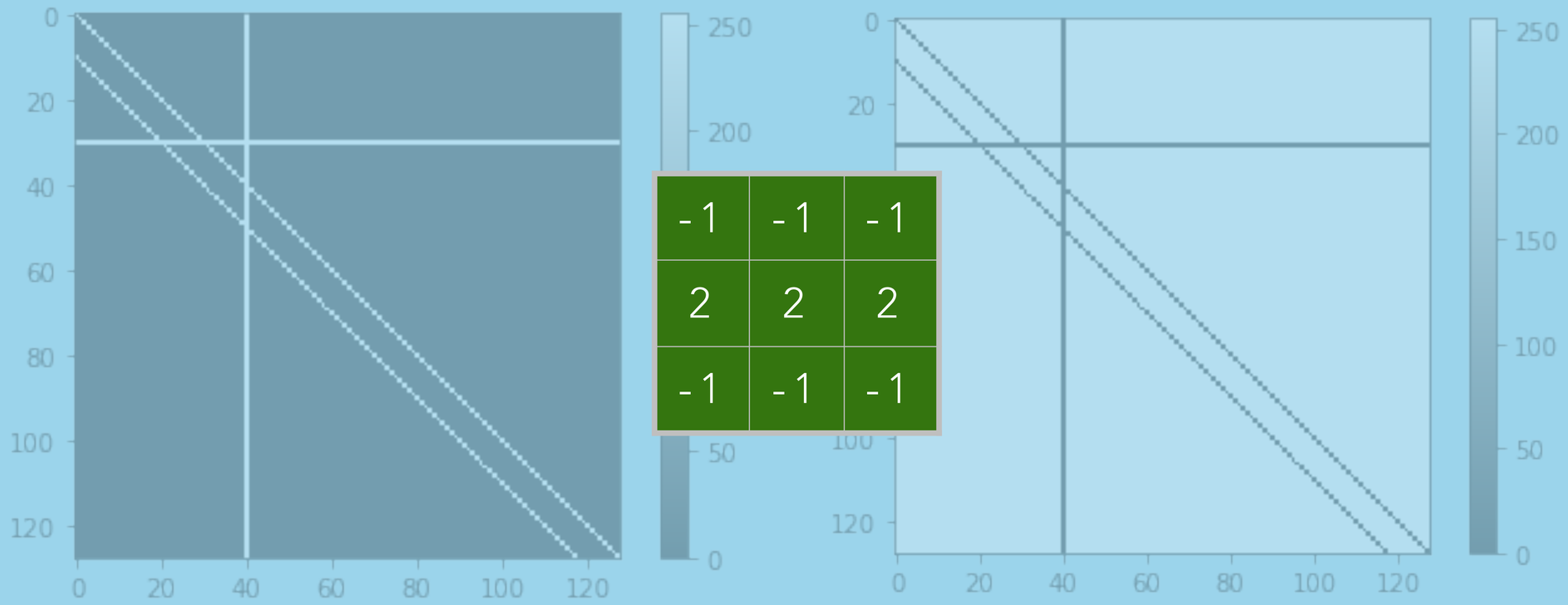
| Filter | Stride | Edges |
| --- | --- | --- |
| (3x3) | 1 | None |



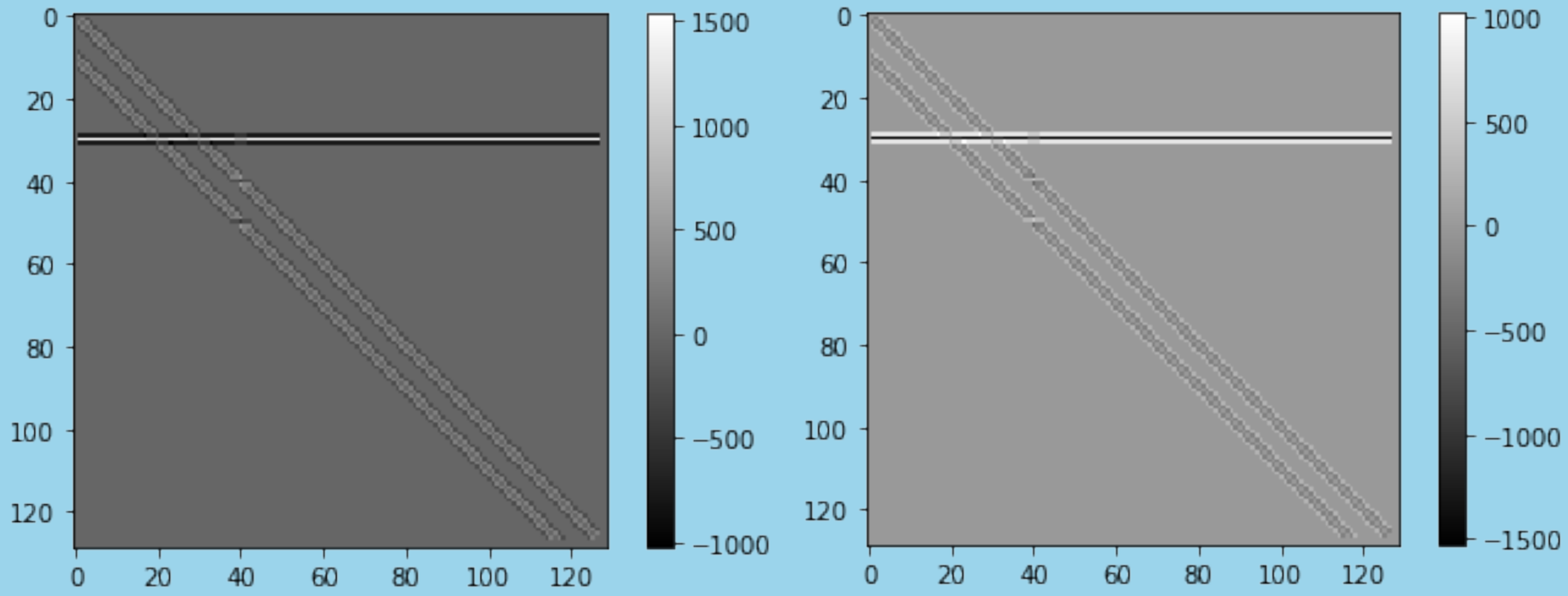
# Discrete Convolutions $C = A \circledast h$

| Filter | Stride | Edges |
|---|---|---|
| (3x3) | 1 | None |



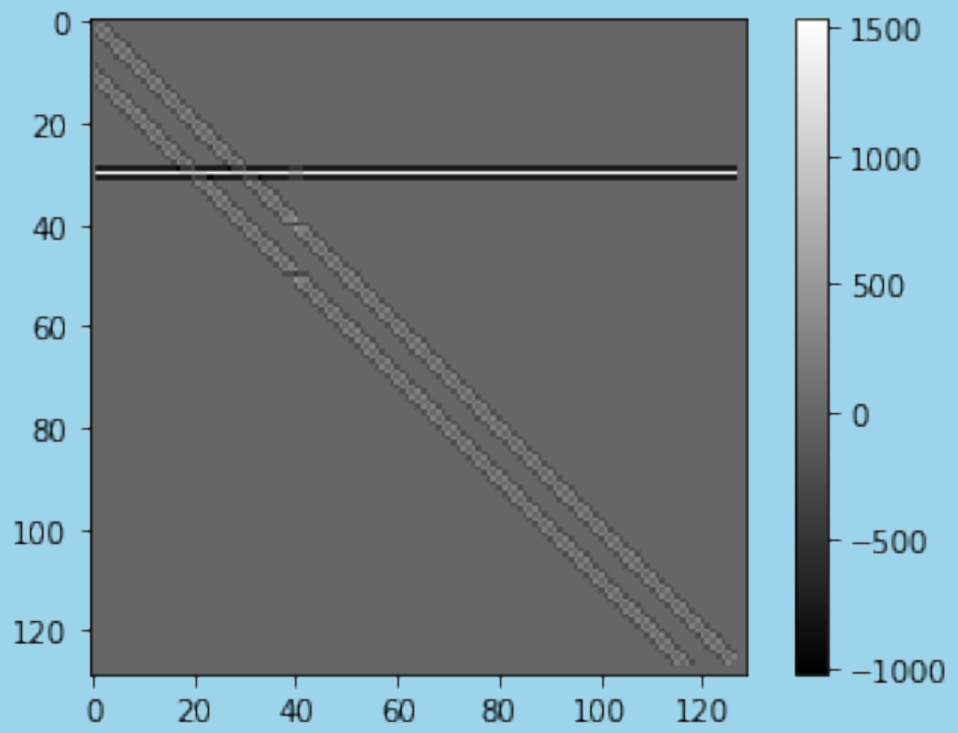
# Discrete Convolutions $C = A \circledast h$

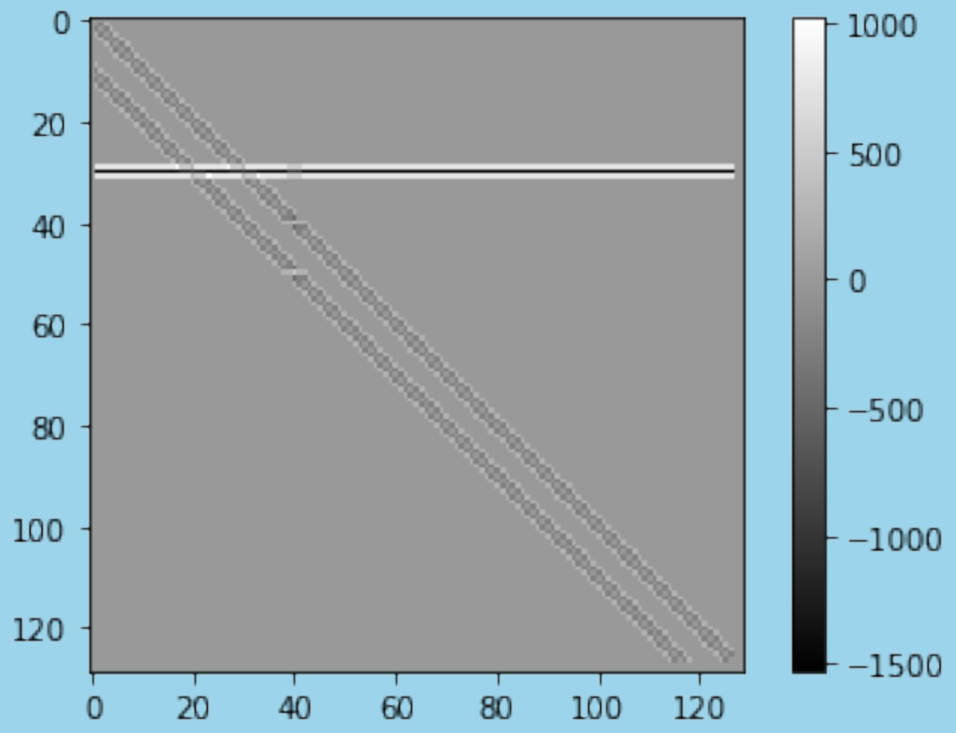
| Filter | Stride | Edges |
| --- | --- | --- |
| (3x3) | 1 | None |



# Discrete Convolutions $C = A \circledast h$

## Line Detection / Extraction



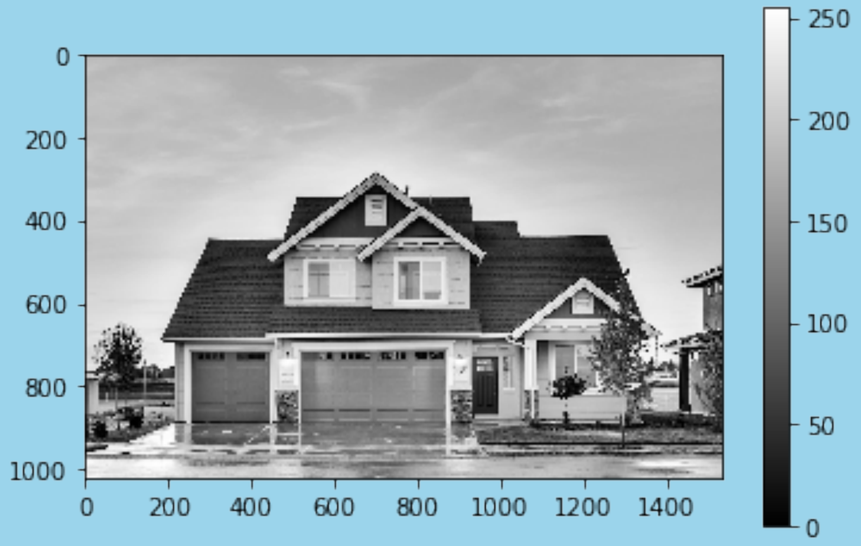
# Discrete Convolutions $C = A \circledast h$

## Photograph

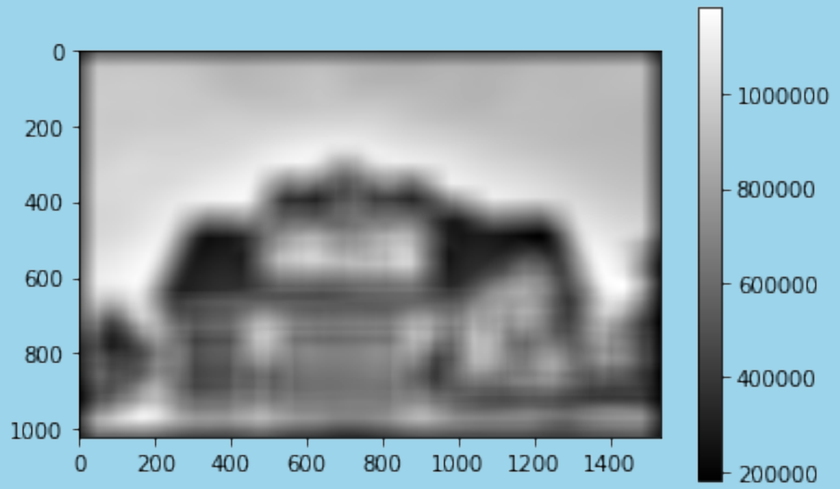
# Discrete Convolutions $C = A \circledast h$

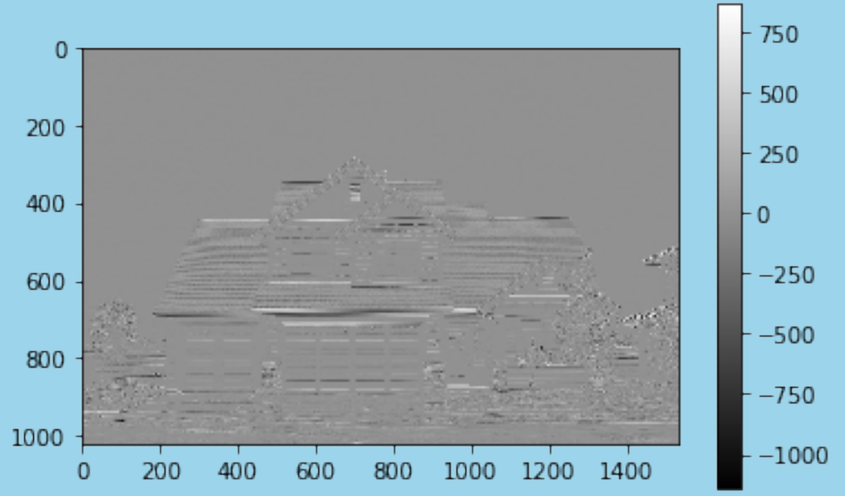
## Photograph

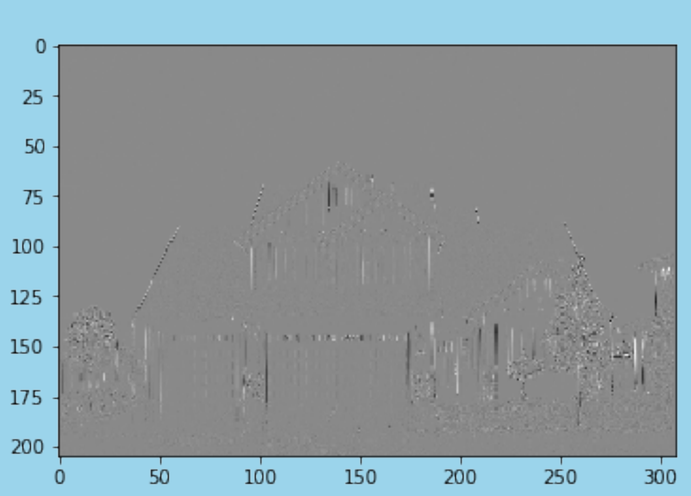
# Discrete Convolutions $C = A \circledast h$




Filter (101x101)

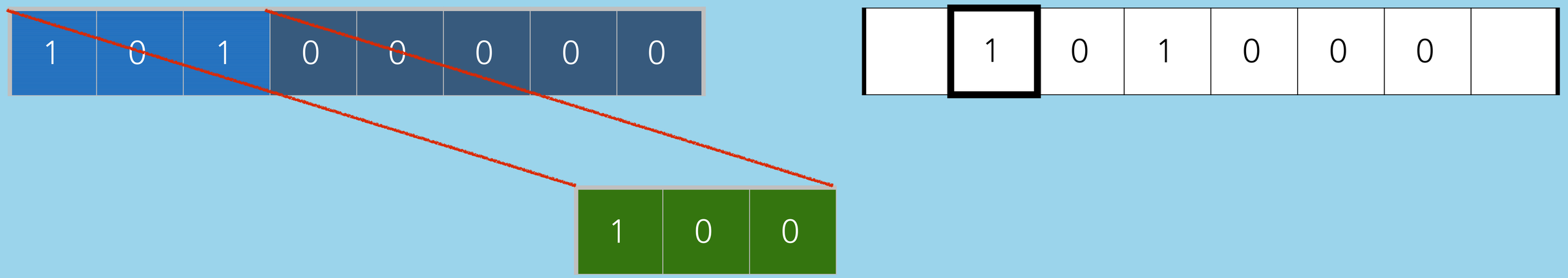


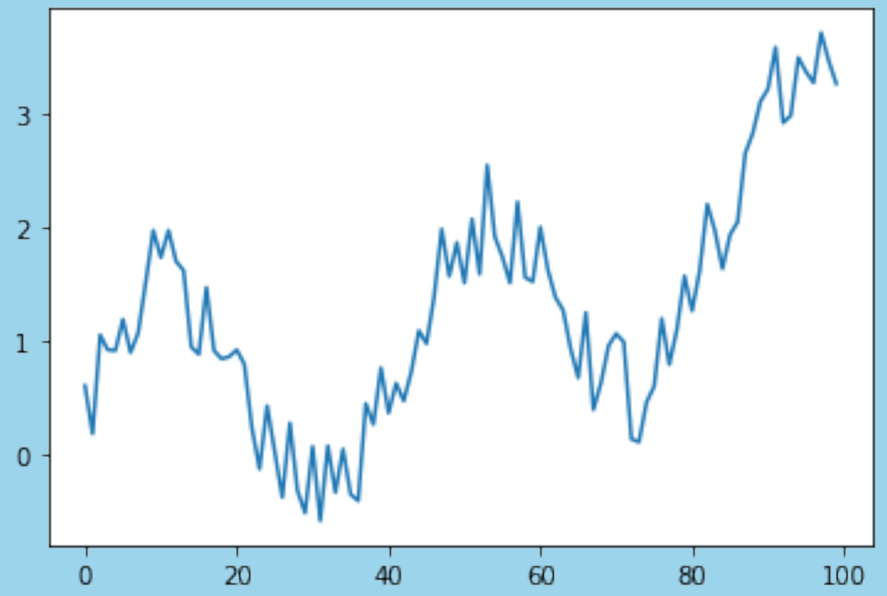
# N-D Discrete Convolutions

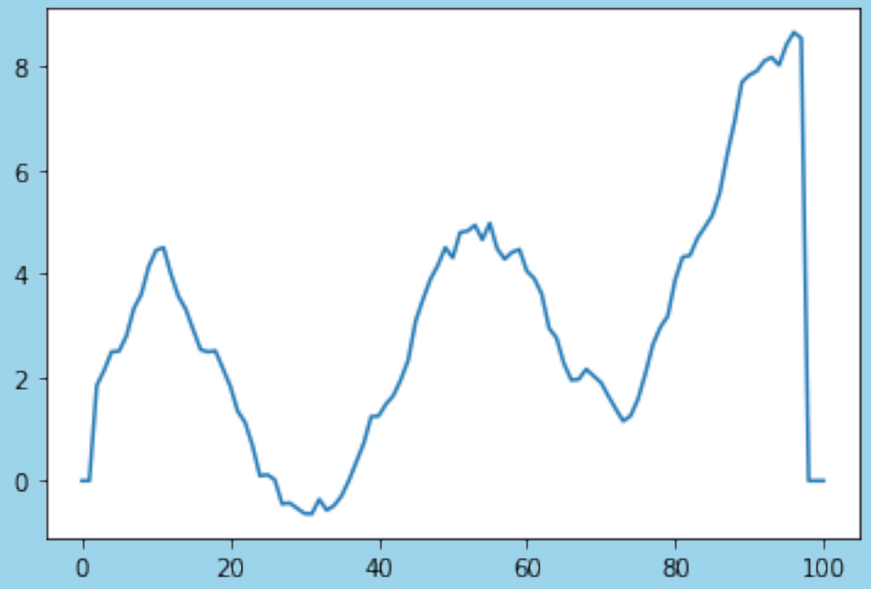
$$C[m,n] = \sum_{i=-\omega}^{\omega} h[i+\omega] * A[m+i]$$



# *N*-D Discrete Convolutions

| Filter | Stride | Edges |
| --- | --- | --- |
| (1x5) | 1 | None |




| | | | | |
| --- | --- | --- | --- | --- |
| 0.5 | 0.5 | 0.5 | 0.5 | 0.5 |

# Discrete Convolutions

Discrete convolutions apply to **array** or **matrix**-like data

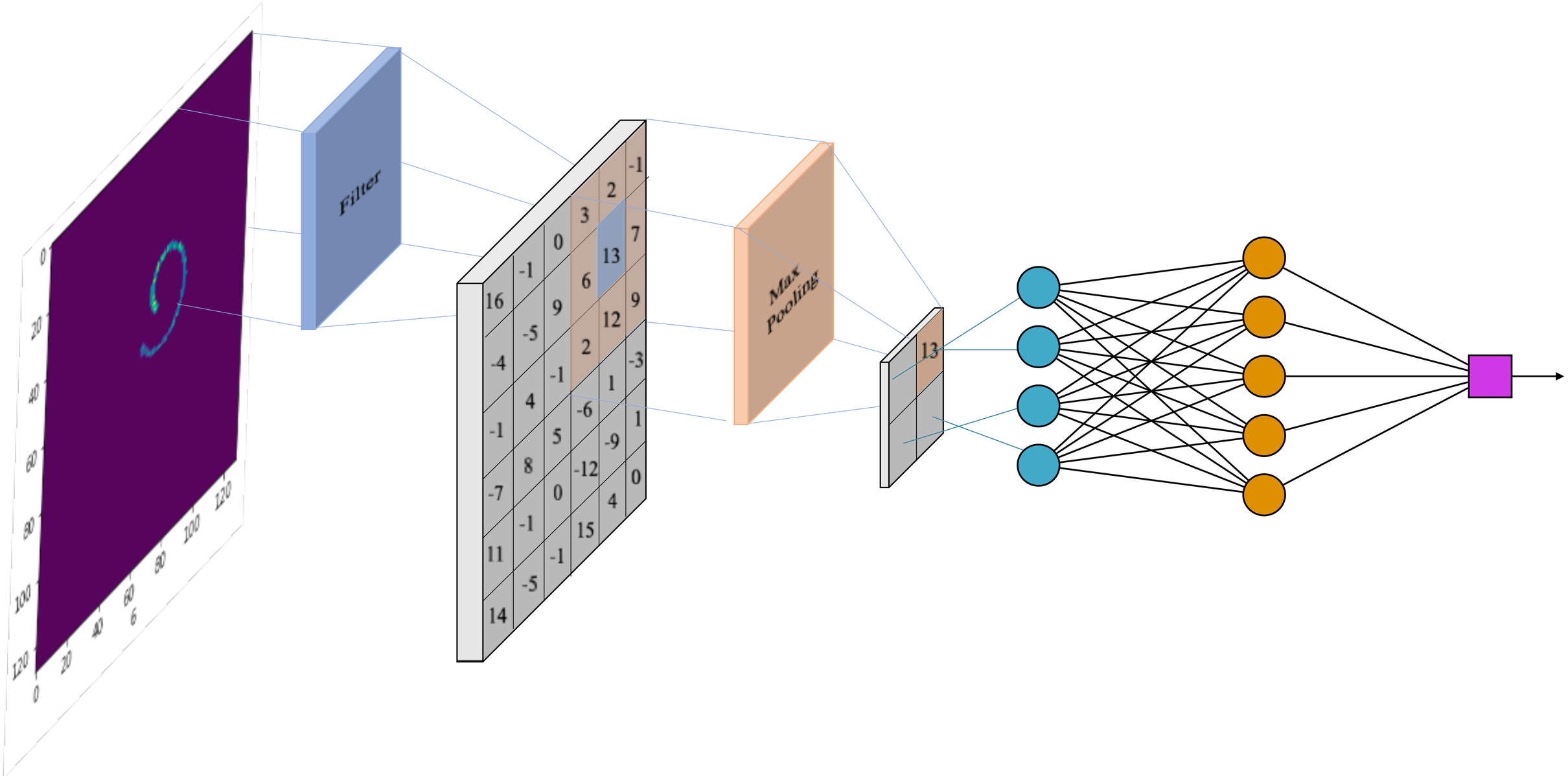
Discrete convolutions are matrix operations that can be used to apply **filters** to a matrix or array

These filters can **extract features** or transform the input

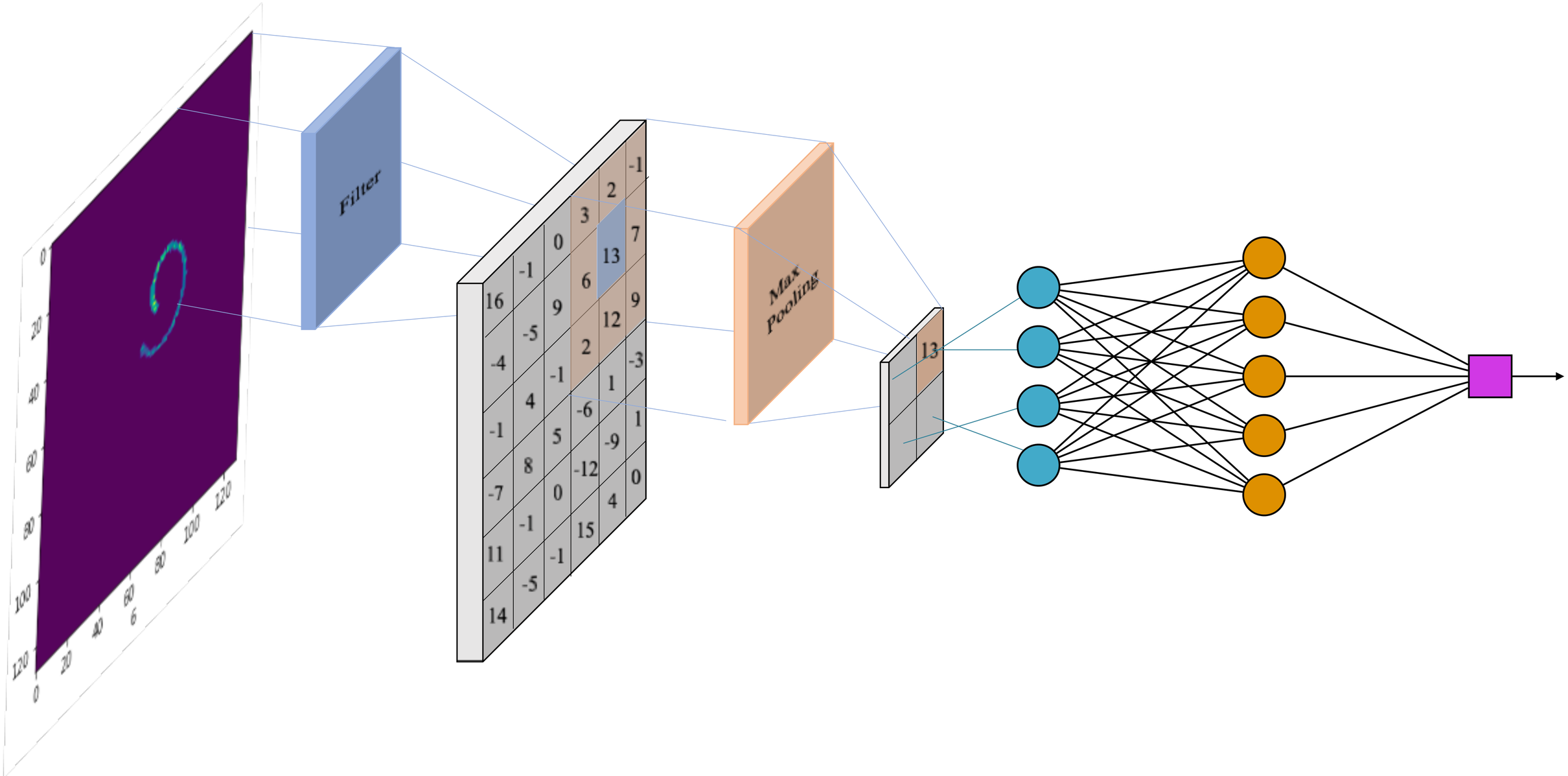
# CONVOLUTIONAL NEURAL NETWORKS:

## ARCHITECTURE AND TRAINING

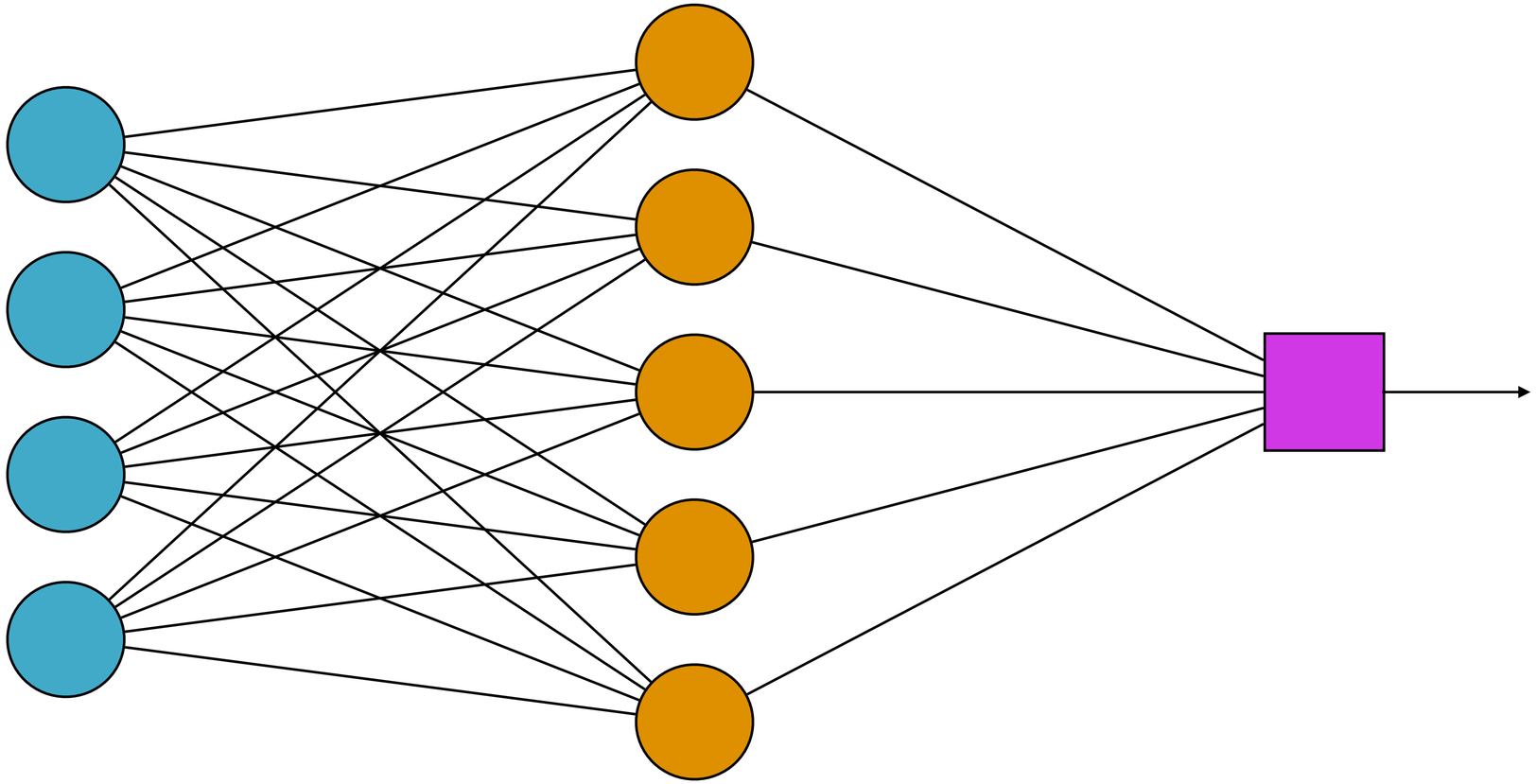
# CONVOLUTIONAL NEURAL NETWORKS

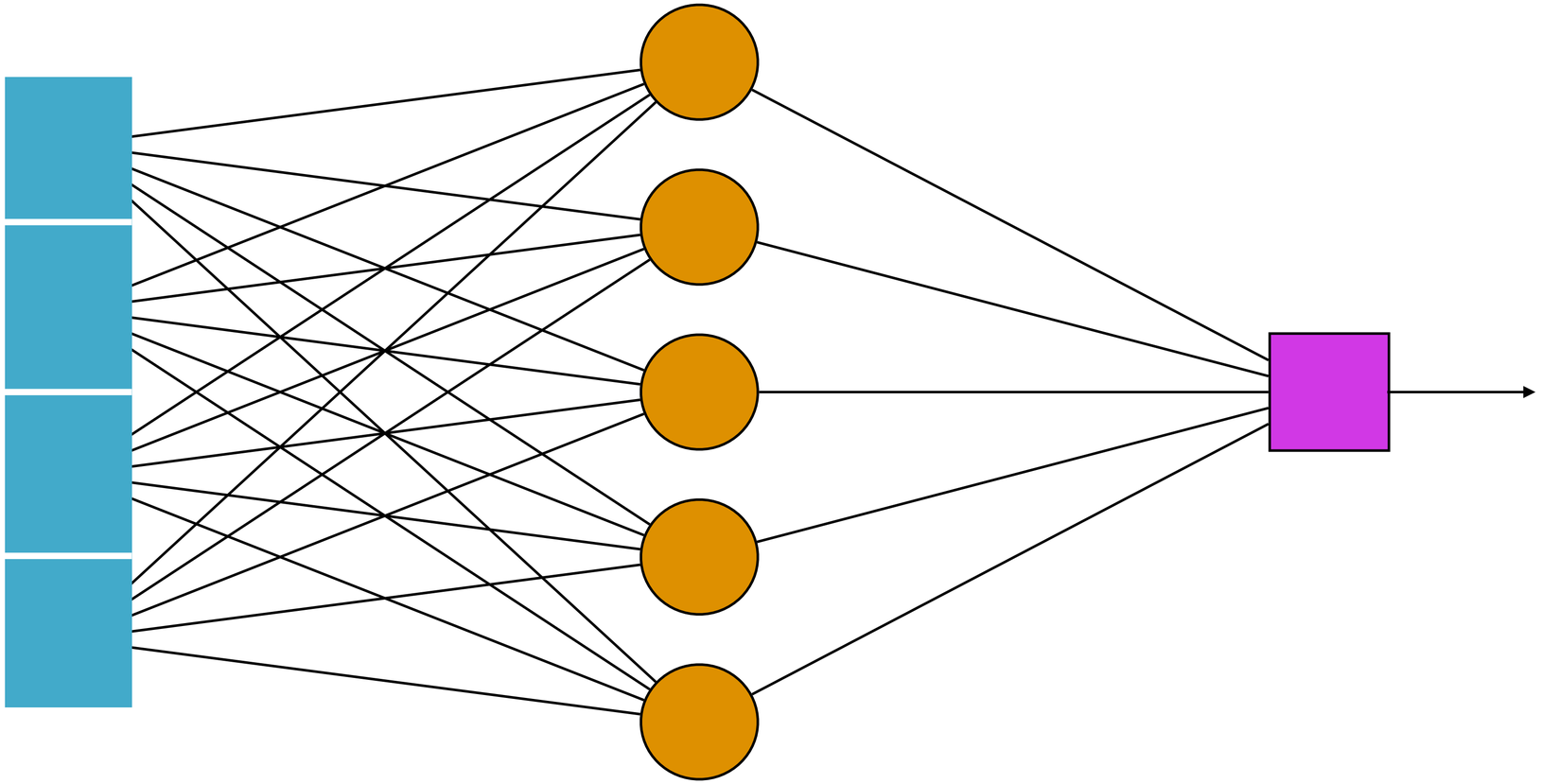


# CHECK: WHY ADD DISCRETE CONVOLUTIONS?



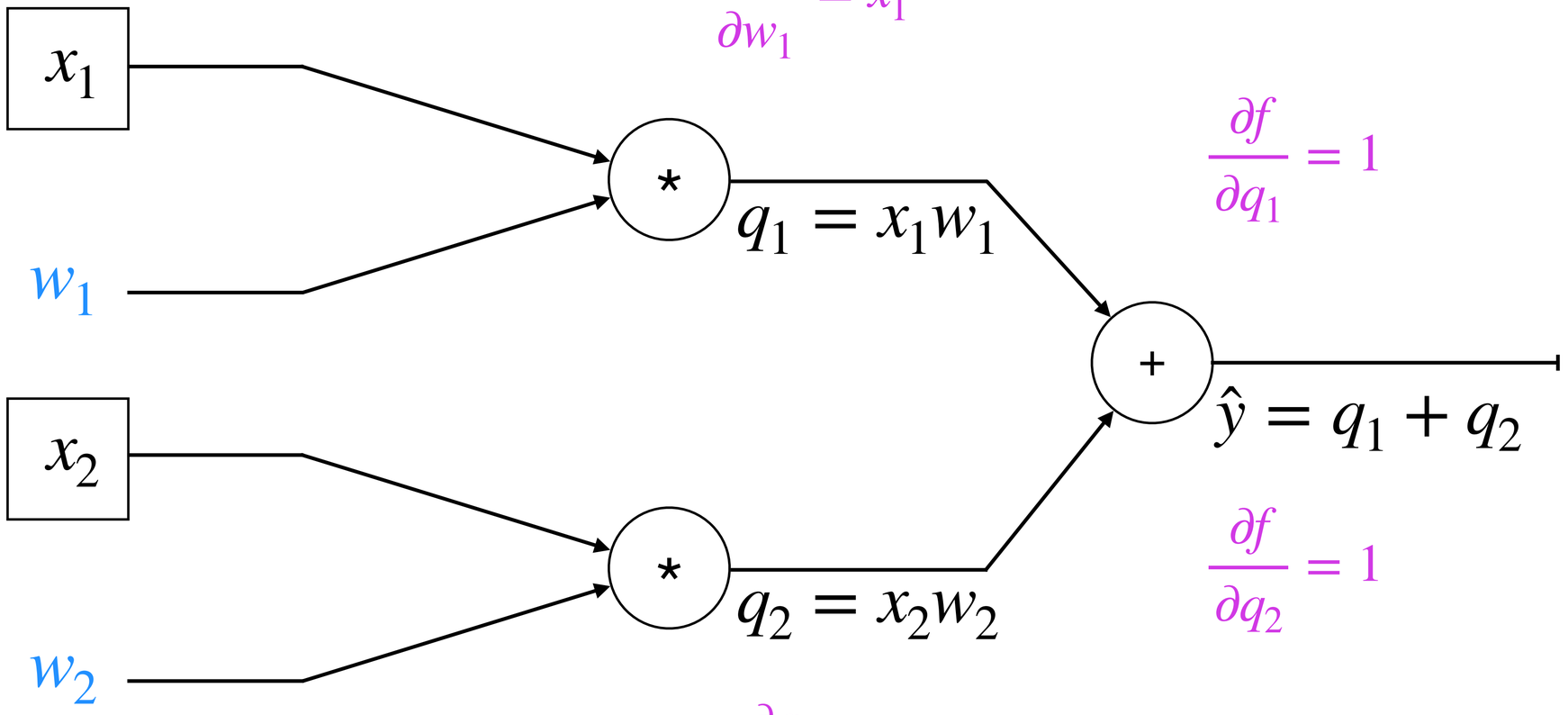
# INPUTS CONSIDERED "UNORDERED"





# BACKPROPAGATION

$$w_1 = w_1 + \eta * \frac{\partial J}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial q_1} \frac{\partial q_1}{\partial w_1}$$




$$\frac{\partial q_1}{\partial w_1} = x_1$$

$$\frac{\partial f}{\partial q_1} = 1$$

$$\frac{\partial J}{\partial \hat{y}} = 1$$

$$\frac{\partial f}{\partial q_2} = 1$$

$$\frac{\partial q_2}{\partial w_2} = x_2$$

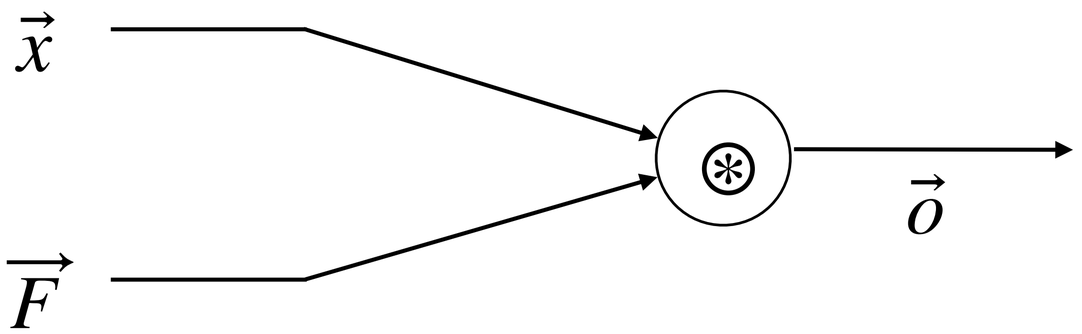
$$w_2 = w_2 + \eta * \frac{\partial J}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial q_1} \frac{\partial q_1}{\partial w_2}$$

Loss function

$$J(w) = \hat{y} - y$$

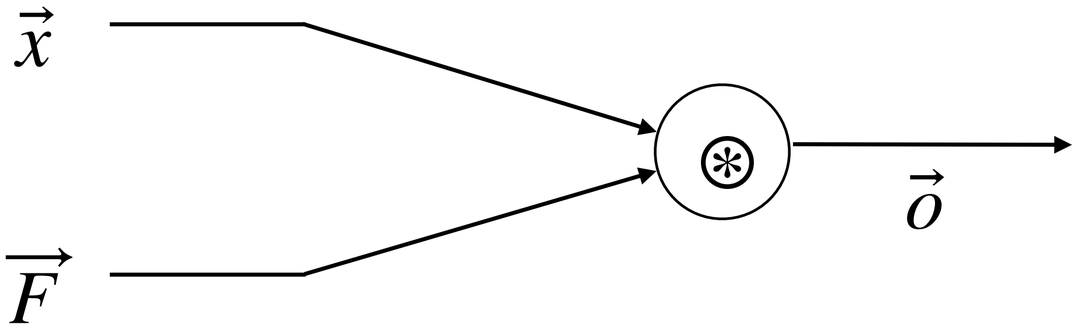
# BACKPROPAGATION

$$\vec{o} = \vec{x} \circledast \overrightarrow{F}$$

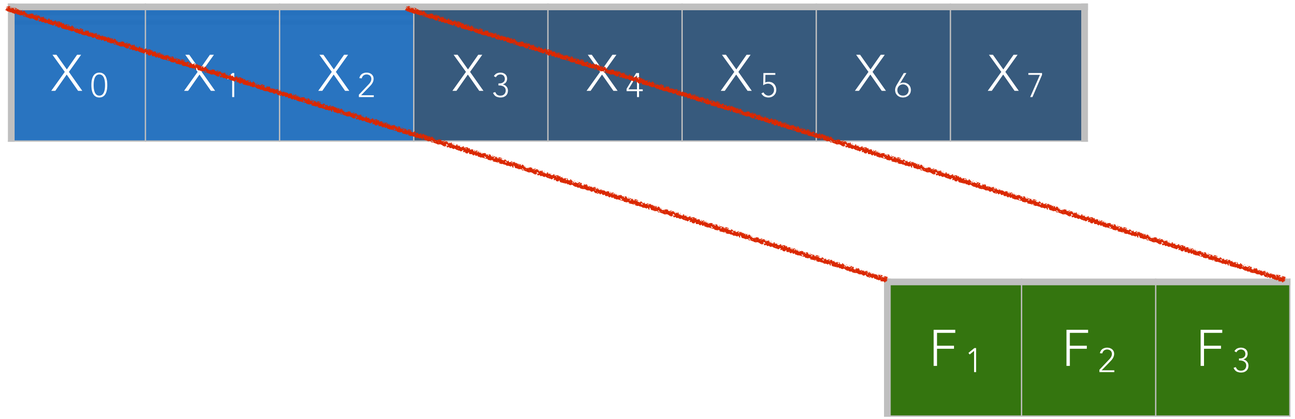


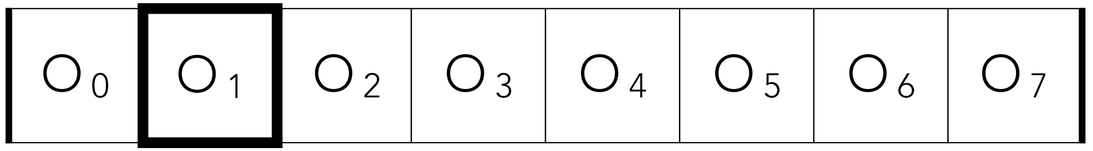
# BACKPROPAGATION

$$\vec{o} = \vec{x} \circledast \vec{F}$$




$$o[n] = (x \circledast F)[n] = \sum_{i=-\omega}^{\omega} x[i+n+\omega] * F[i+\omega]$$



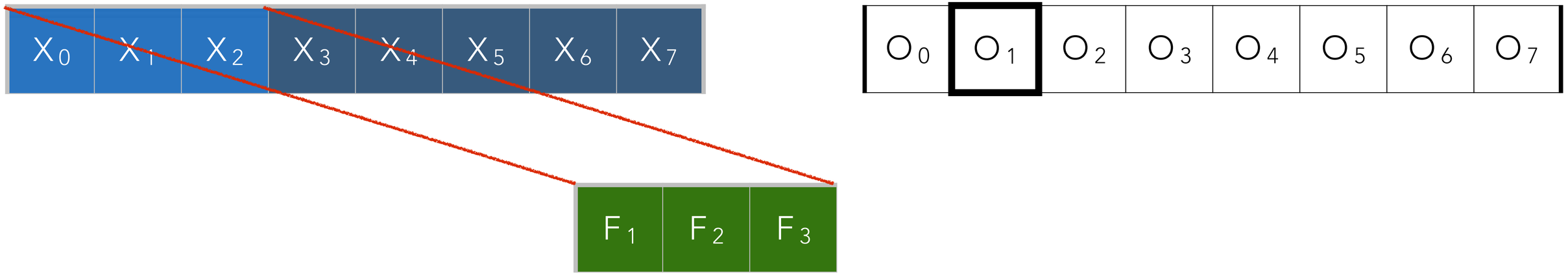



# BACKPROPAGATION

$$F_i \leftarrow F_i - \eta \frac{\partial J}{\partial F_i}$$

$$\frac{\partial J}{\partial F_i} = \sum_{k=1}^{M} \frac{\partial J}{\partial o_k} \frac{\partial o_k}{\partial F_i}$$

$$o[n] = (x \circledast F)[n] = \sum_{i=-\omega}^{\omega} x[i+n+\omega] * F[i+\omega]$$

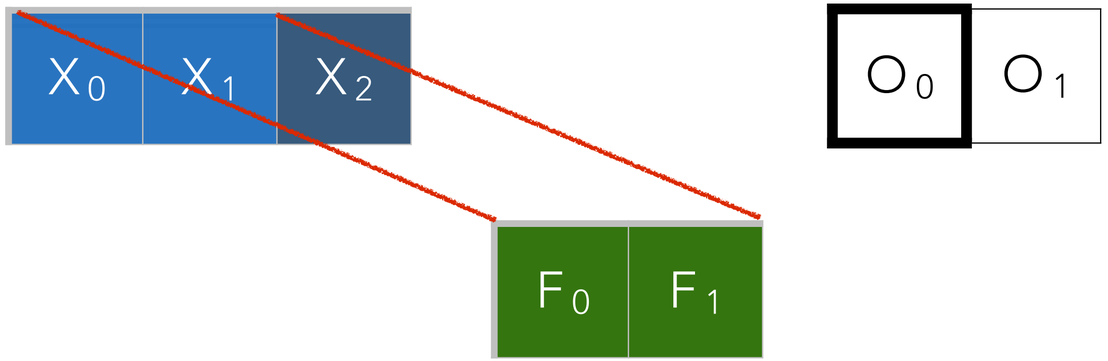


# BACKPROPAGATION

$$F_i \leftarrow F_i - \eta \frac{\partial J}{\partial F_i}$$

$$\frac{\partial J}{\partial F_i} = \sum_{k=1}^{M} \frac{\partial J}{\partial o_k} \frac{\partial o_k}{\partial F_i}$$

$$o[n] = (x \circledast F)[n] = \sum_{i=-\omega}^{\omega} x[i+n+\omega] * F[i+\omega]$$



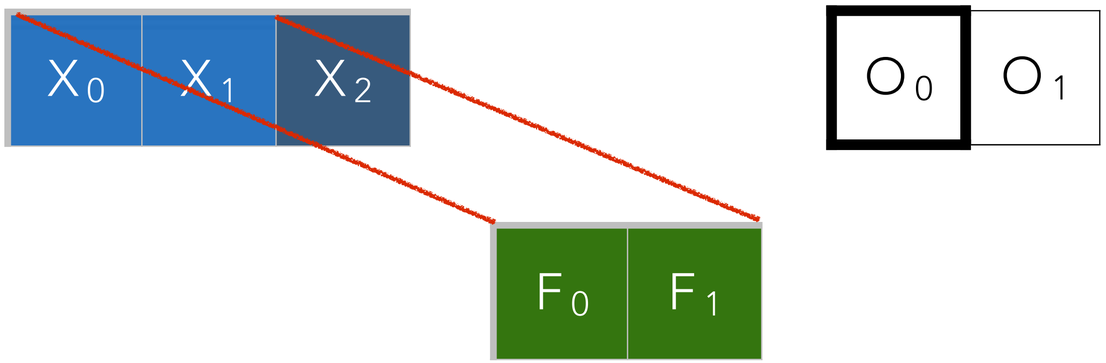
# BACKPROPAGATION

$$\frac{\partial J}{\partial F_1} = \frac{\partial J}{\partial o_0}\frac{\partial o_0}{\partial F_1} + \frac{\partial J}{\partial o_1}\frac{\partial o_1}{\partial F_1}$$

$$F_i \leftarrow F_i - \eta\frac{\partial J}{\partial F_i}$$

$$\frac{\partial J}{\partial F_i} = \sum_{k=1}^{M}\frac{\partial J}{\partial o_k}\frac{\partial o_k}{\partial F_i}$$

$$o[n] = (x \circledast F)[n] = \sum_{i=-\omega}^{\omega} x[i+n+\omega] * F[i+\omega]$$



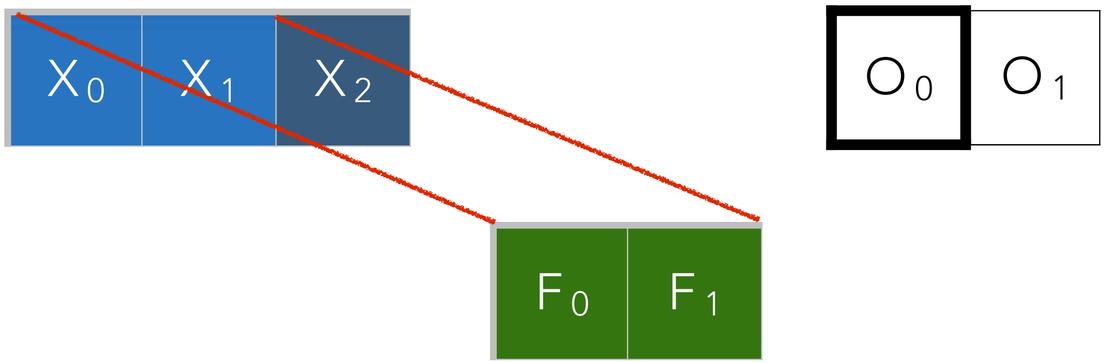
# BACKPROPAGATION

$$\frac{\partial J}{\partial F_1} = \frac{\partial J}{\partial o_0}\frac{\partial o_0}{\partial F_1} + \frac{\partial J}{\partial o_1}\frac{\partial o_1}{\partial F_1}$$

$$F_i \leftarrow F_i - \eta\frac{\partial J}{\partial F_i}$$

$$\frac{\partial J}{\partial F_i} = \sum_{k=1}^{M}\frac{\partial J}{\partial o_k}\frac{\partial o_k}{\partial F_i}$$

$$o[n] = (x \circledast F)[n] = \sum_{i=-\omega}^{\omega} x[i+n+\omega] * F[i+\omega]$$



# BACKPROPAGATION

$$\frac{\partial J}{\partial F_1} = \frac{\partial J}{\partial o_0}\frac{\partial o_0}{\partial F_1} + \frac{\partial J}{\partial o_1}\frac{\partial o_1}{\partial F_1}$$

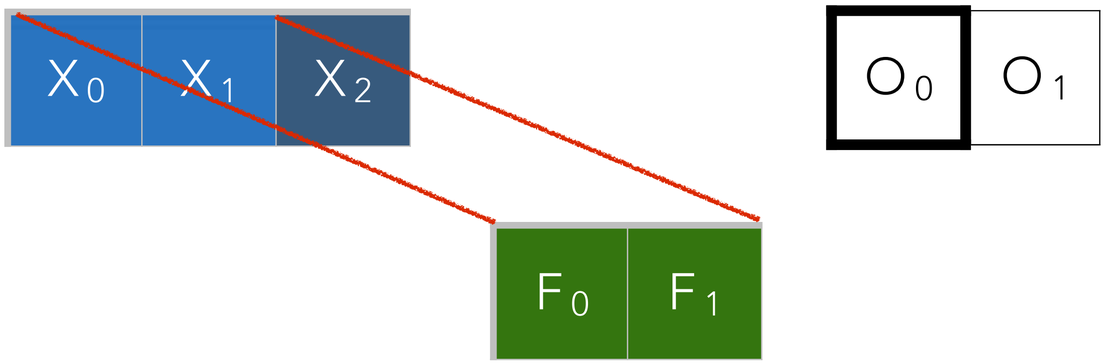
$$o_0 = F_0 x_0 + F_1 x_1$$

$$o_1 = F_0 x_1 + F_1 x_2$$

$$F_i \leftarrow F_i - \eta \frac{\partial J}{\partial F_i}$$

$$\frac{\partial J}{\partial F_i} = \sum_{k=1}^{M} \frac{\partial J}{\partial o_k}\frac{\partial o_k}{\partial F_i}$$

$$o[n] = (x \circledast F)[n] = \sum_{i=-\omega}^{\omega} x[i+n+\omega] * F[i+\omega]$$



# BACKPROPAGATION

$$\frac{\partial J}{\partial F_1} = \frac{\partial J}{\partial o_0}\frac{\partial o_0}{\partial F_1} + \frac{\partial J}{\partial o_1}\frac{\partial o_1}{\partial F_1}$$

$$F_i \leftarrow F_i - \eta \frac{\partial J}{\partial F_i}$$

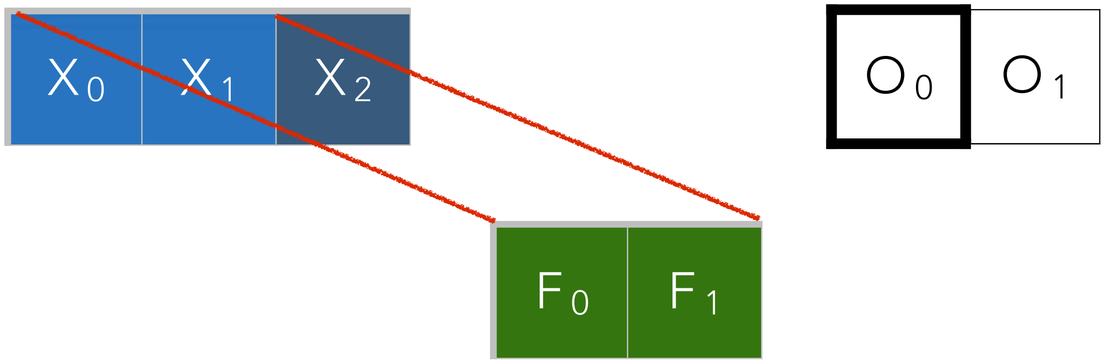
$$\frac{\partial J}{\partial F_i} = \sum_{k=1}^{M} \frac{\partial J}{\partial o_k}\frac{\partial o_k}{\partial F_i}$$

$$o_0 = F_0 x_0 + F_1 x_1$$
$$o_1 = F_0 x_1 + F_1 x_2$$

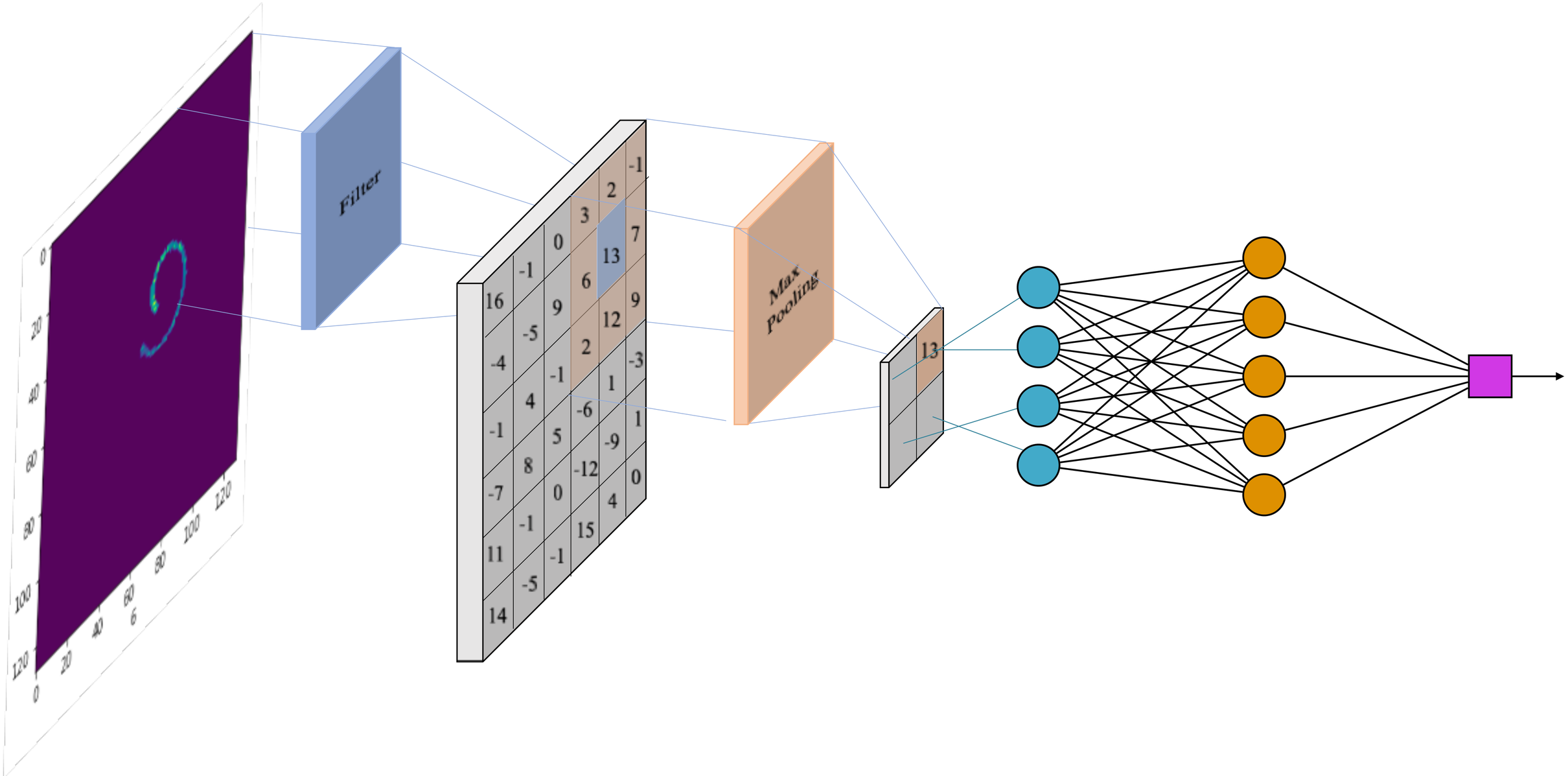
$$o[n] = (x \circledast F)[n] = \sum_{i=-\omega}^{\omega} x[i+n+\omega] * F[i+\omega]$$

$$\frac{\partial o_0}{\partial F_1} = x_1$$

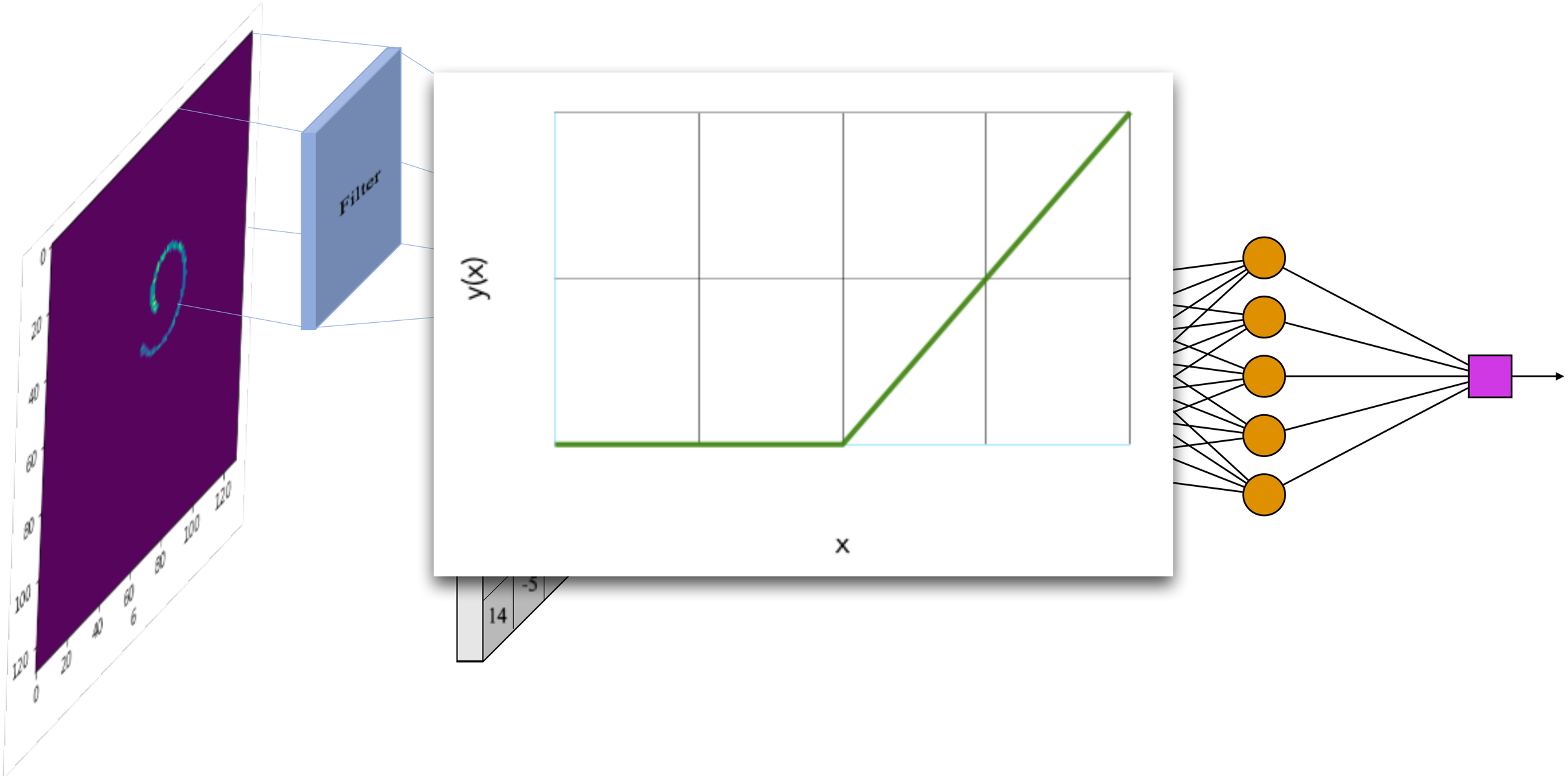
$$\frac{\partial o_1}{\partial F_1} = x_2$$



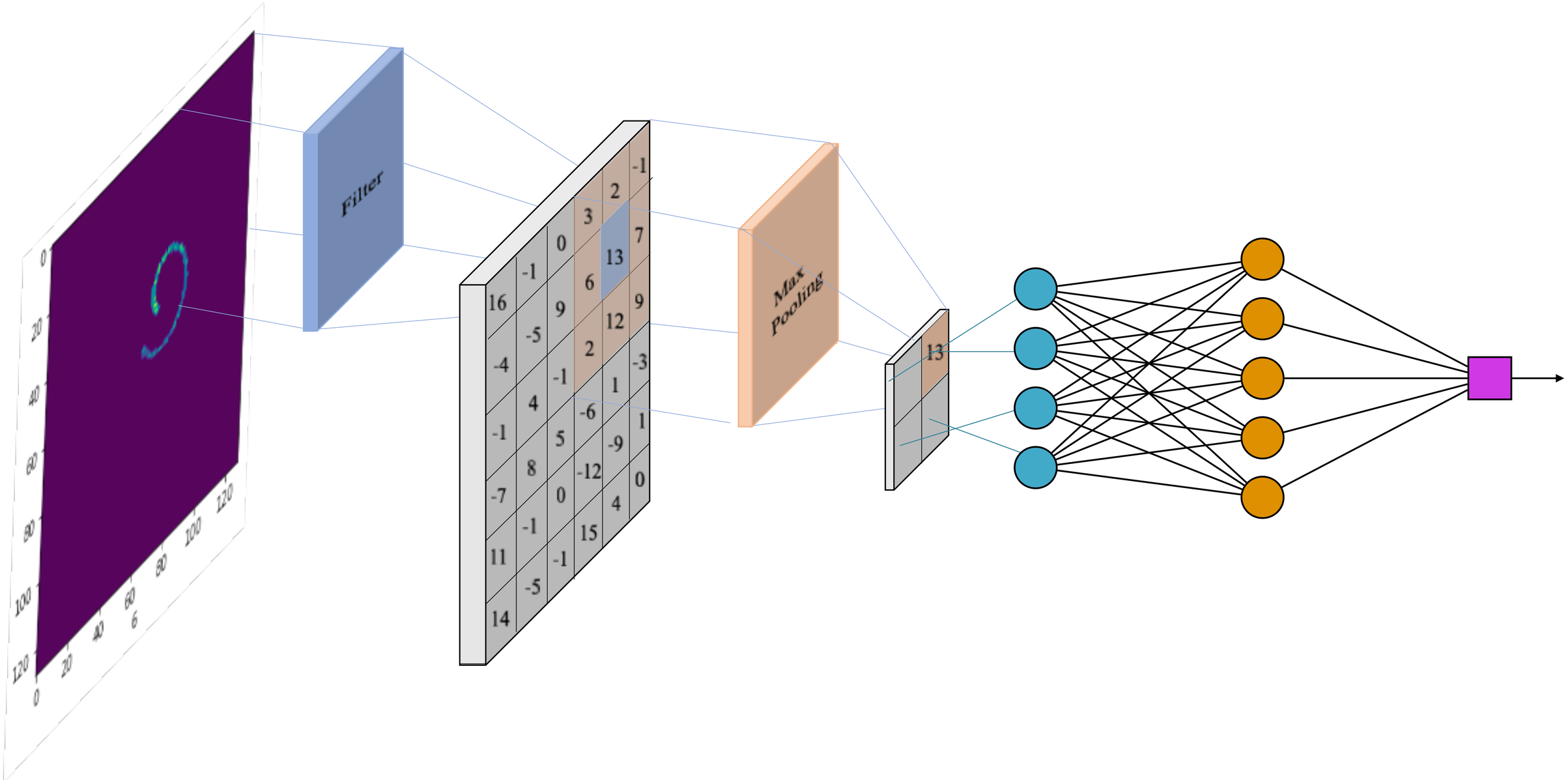
# CONVOLUTIONAL NEURAL NETWORKS



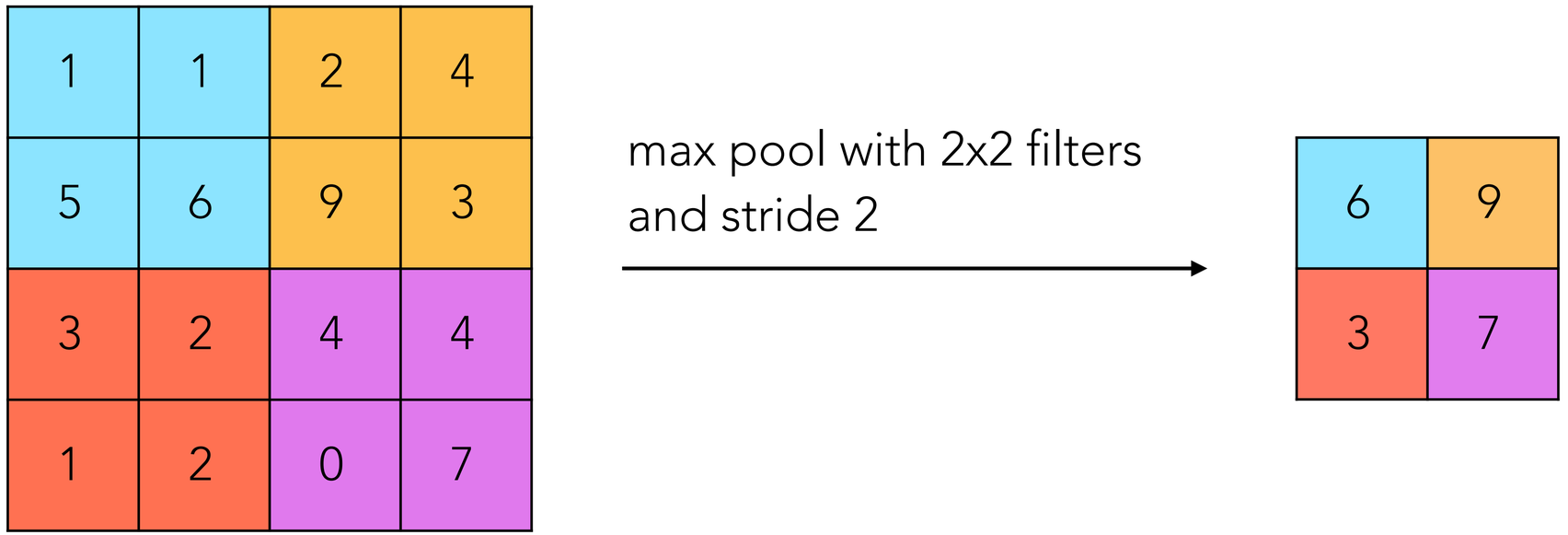
# CONVOLUTIONAL NEURAL NETWORKS



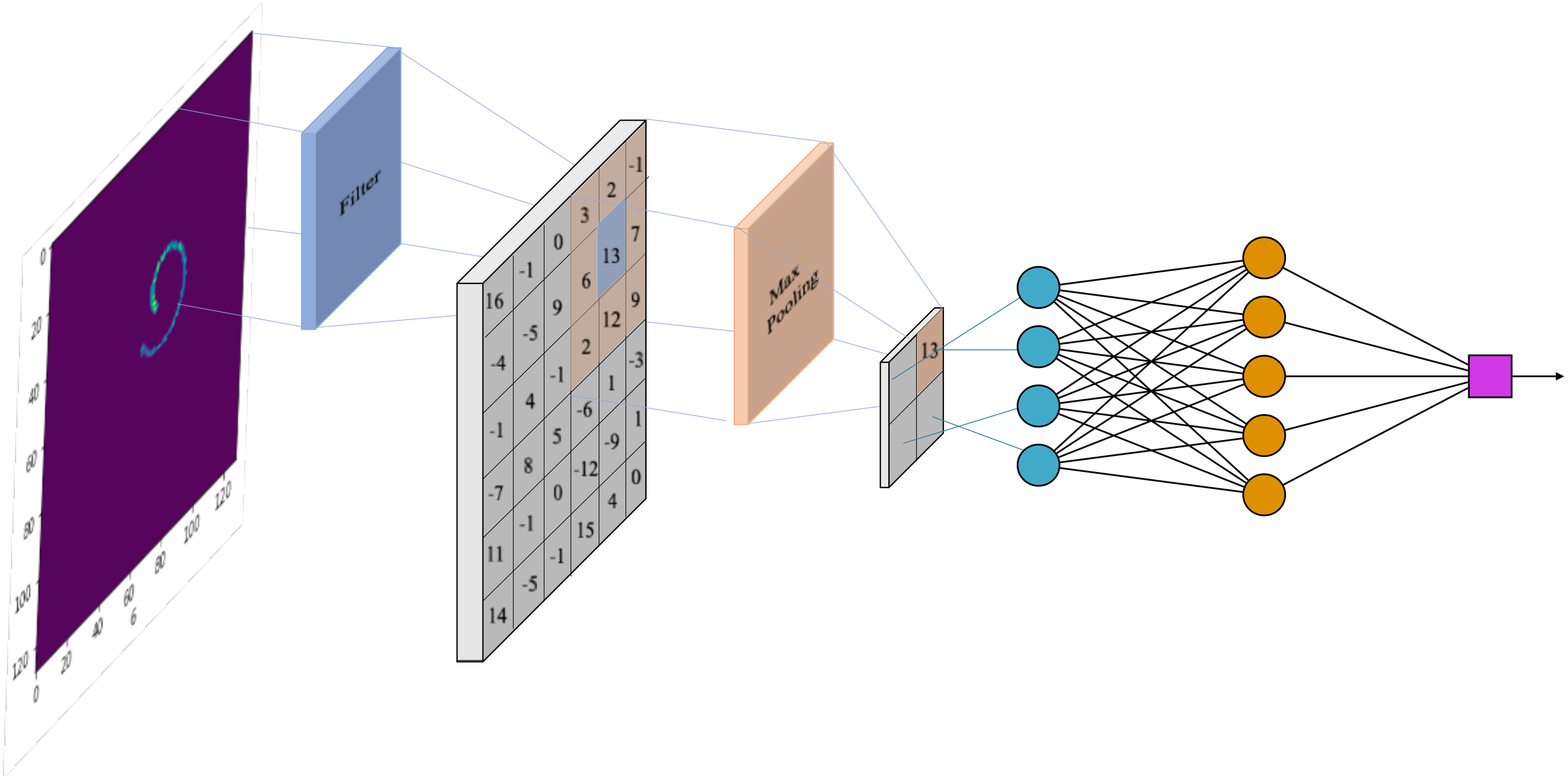
# CONVOLUTIONAL NEURAL NETWORKS



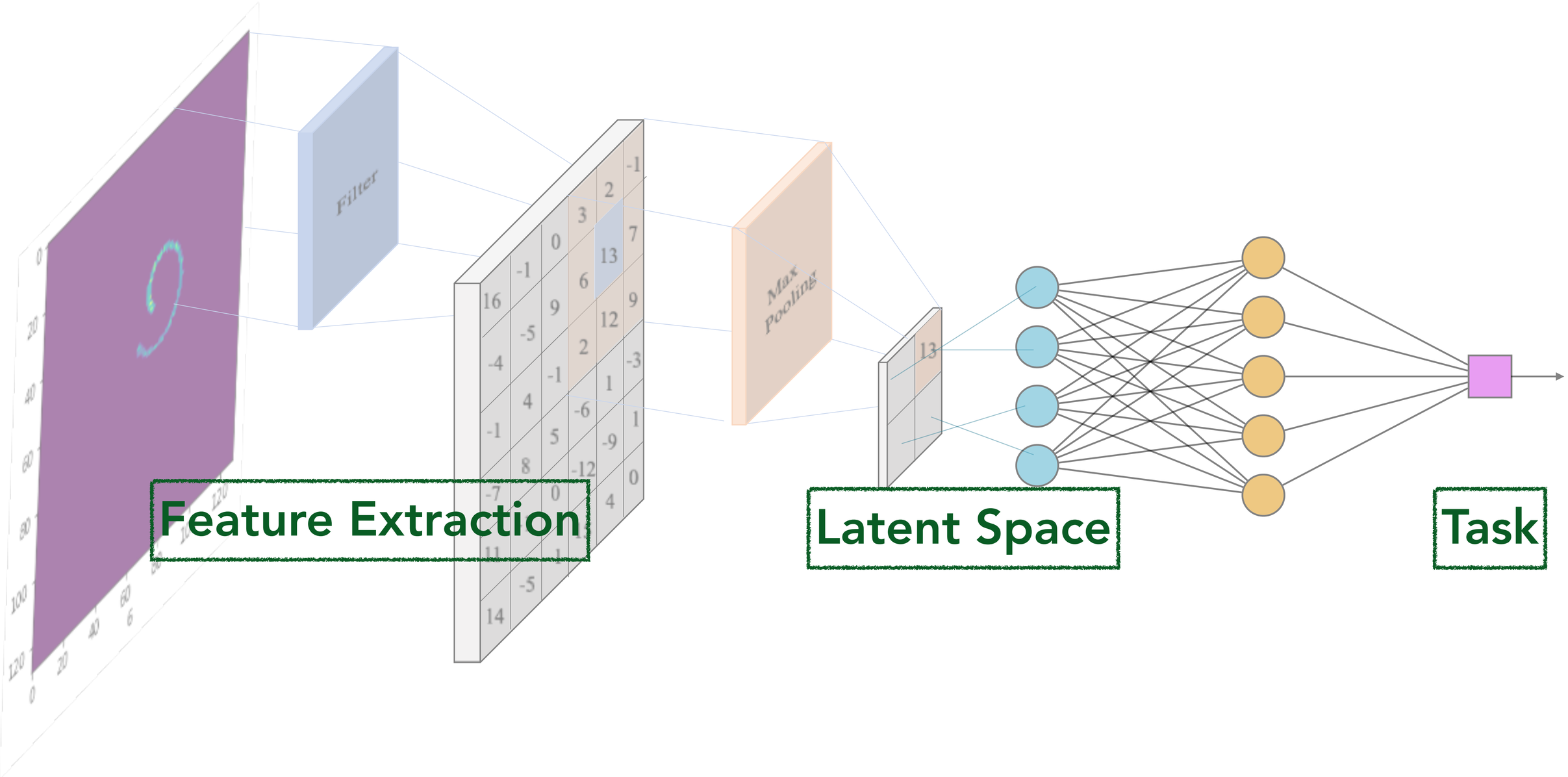
# POOLING


1 1 2 4
5 6 9 3
3 2 4 4
1 2 0 7
max pool with 2x2 filters
and stride 2
6 9
3 7

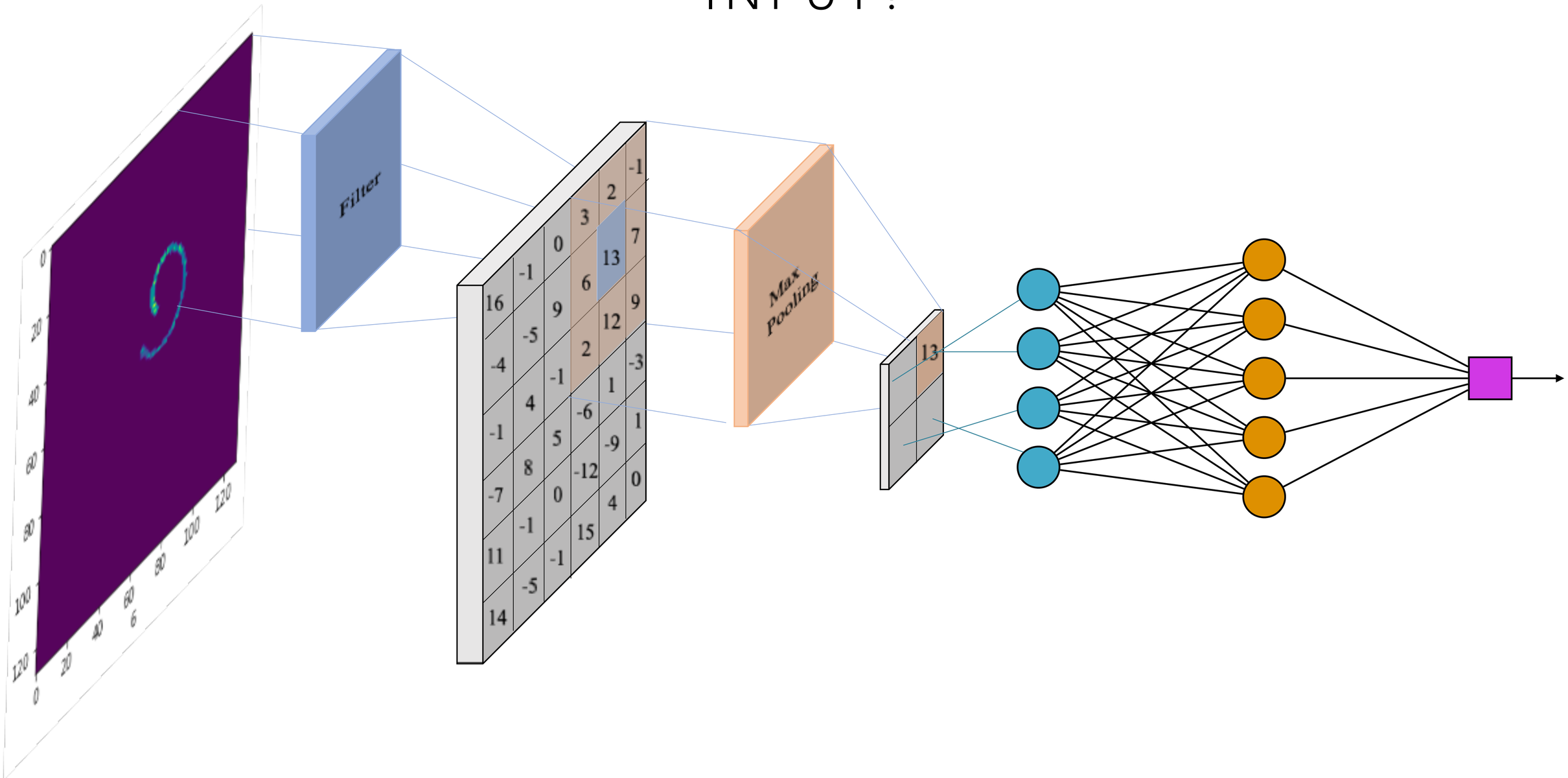
# BACKPROPOGATION



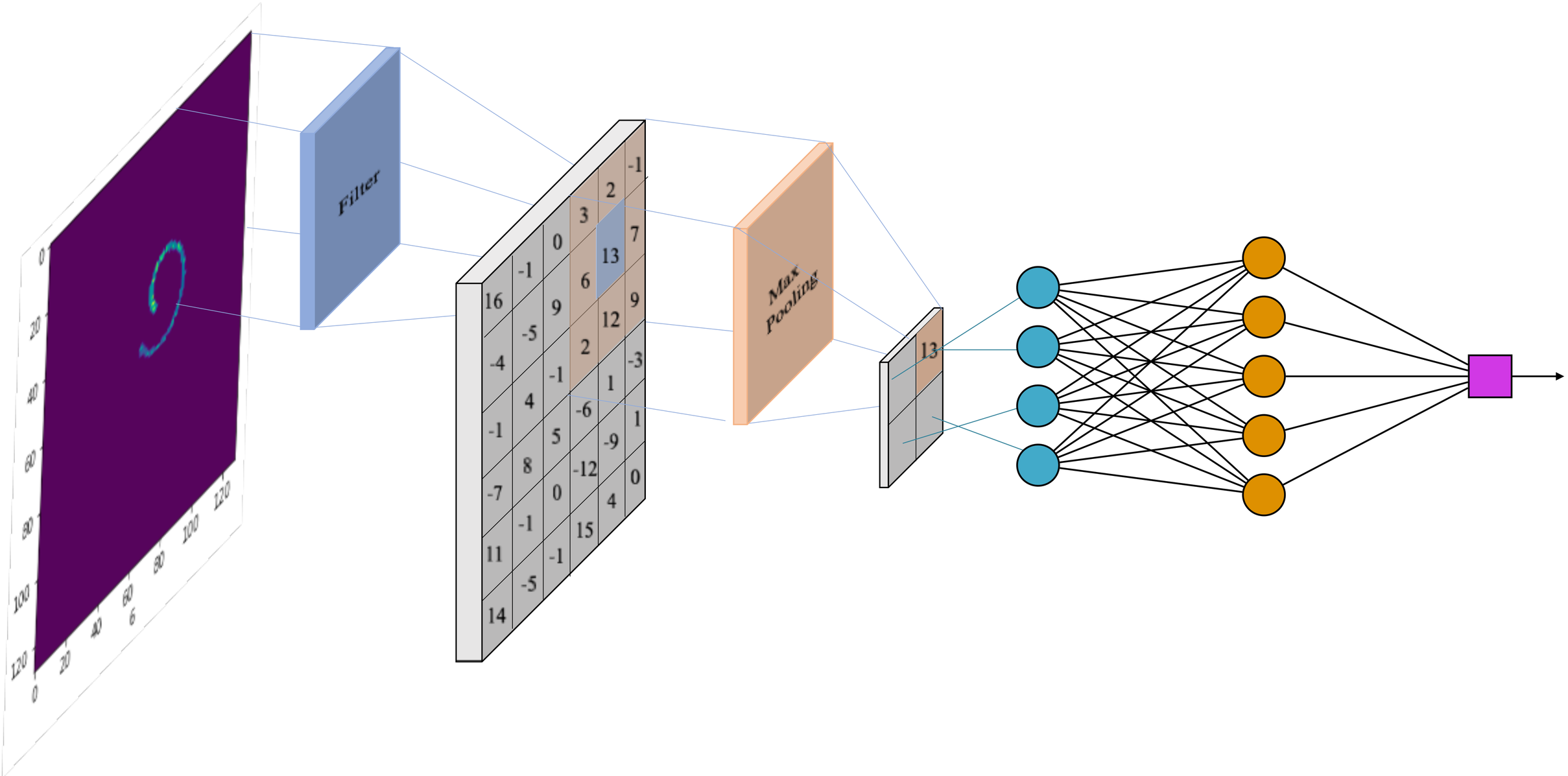
# CONVOLUTIONAL NEURAL NETWORKS



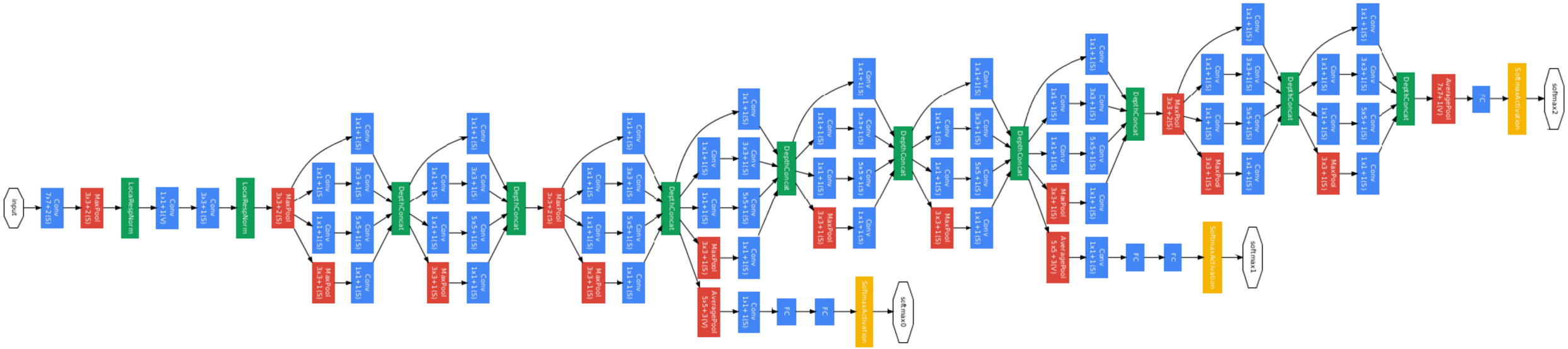
# CHECK: WHAT HAPPENS WITH TRANSLATION IN INPUT?



# CHECK: PROS? CONS?



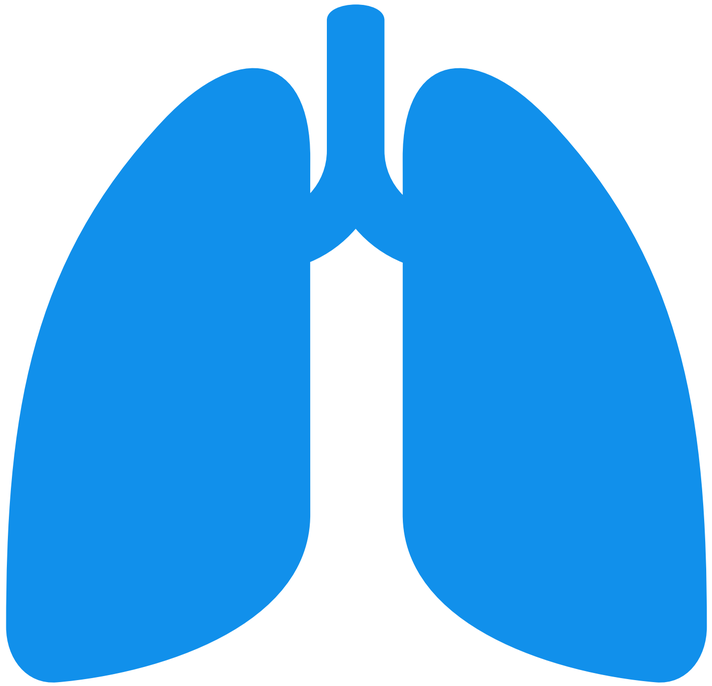
# CONVOLUTIONAL NEURAL NETWORKS




CHRISTIAN SZEGEDY ET. AL. GOING DEEPER WITH CONVOLUTIONS.

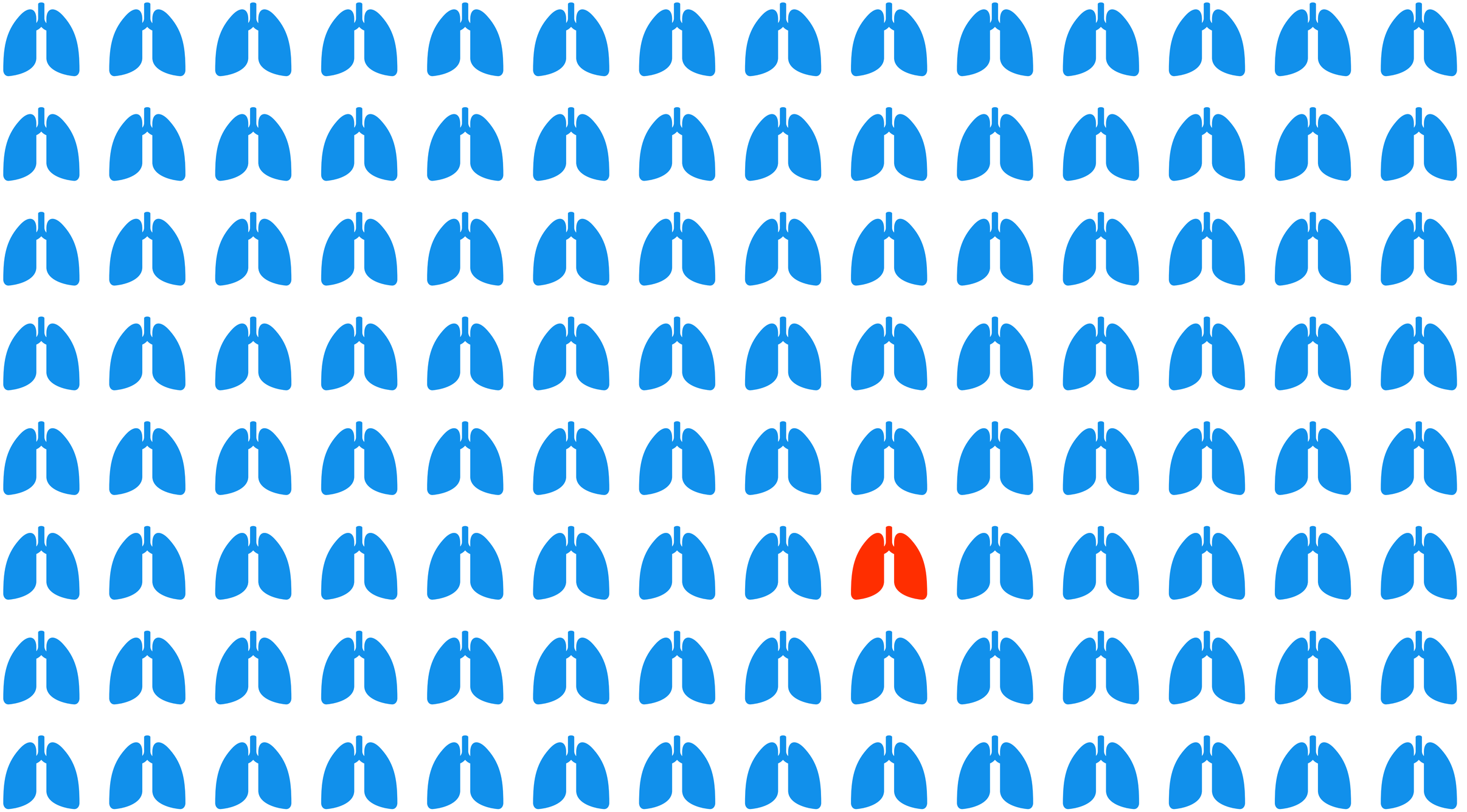
**Application 2:** Can we use machine learning to **accurately** classify events in detectors?

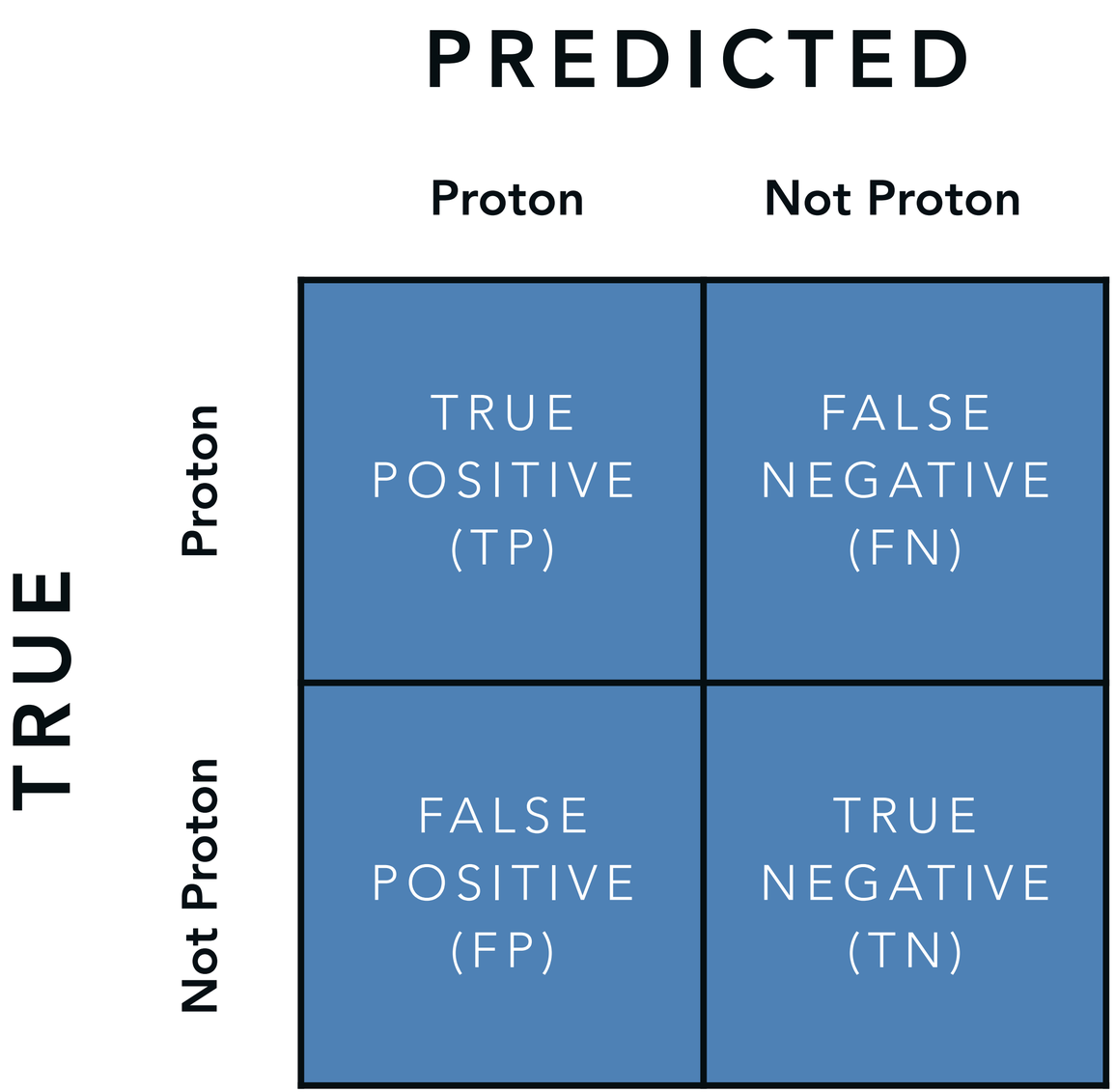
**Metrics**

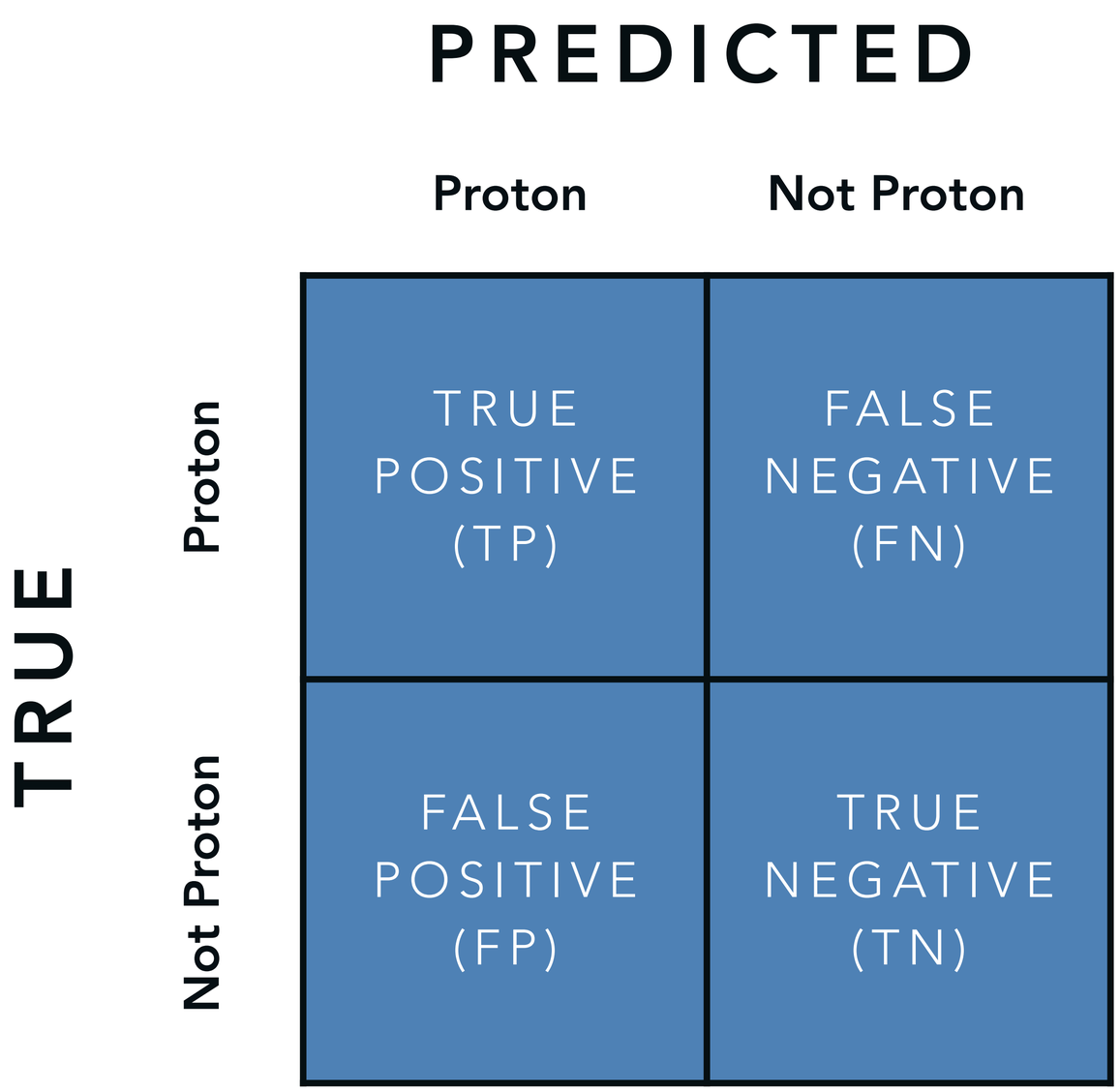


Detect Lung Cancer

99% Accuracy




PREDICTED
Proton
Not Proton
TRUE
Proton
Not Proton
TRUE POSITIVE (TP)
FALSE NEGATIVE (FN)
FALSE POSITIVE (FP)
TRUE NEGATIVE (TN)




$$\text{accuracy} = \frac{TP + TN}{TP + FN + FP + TN}$$

$$\text{precision} = \frac{TP}{TP + FP}$$

$$\text{recall} = \frac{TP}{TP + FN}$$

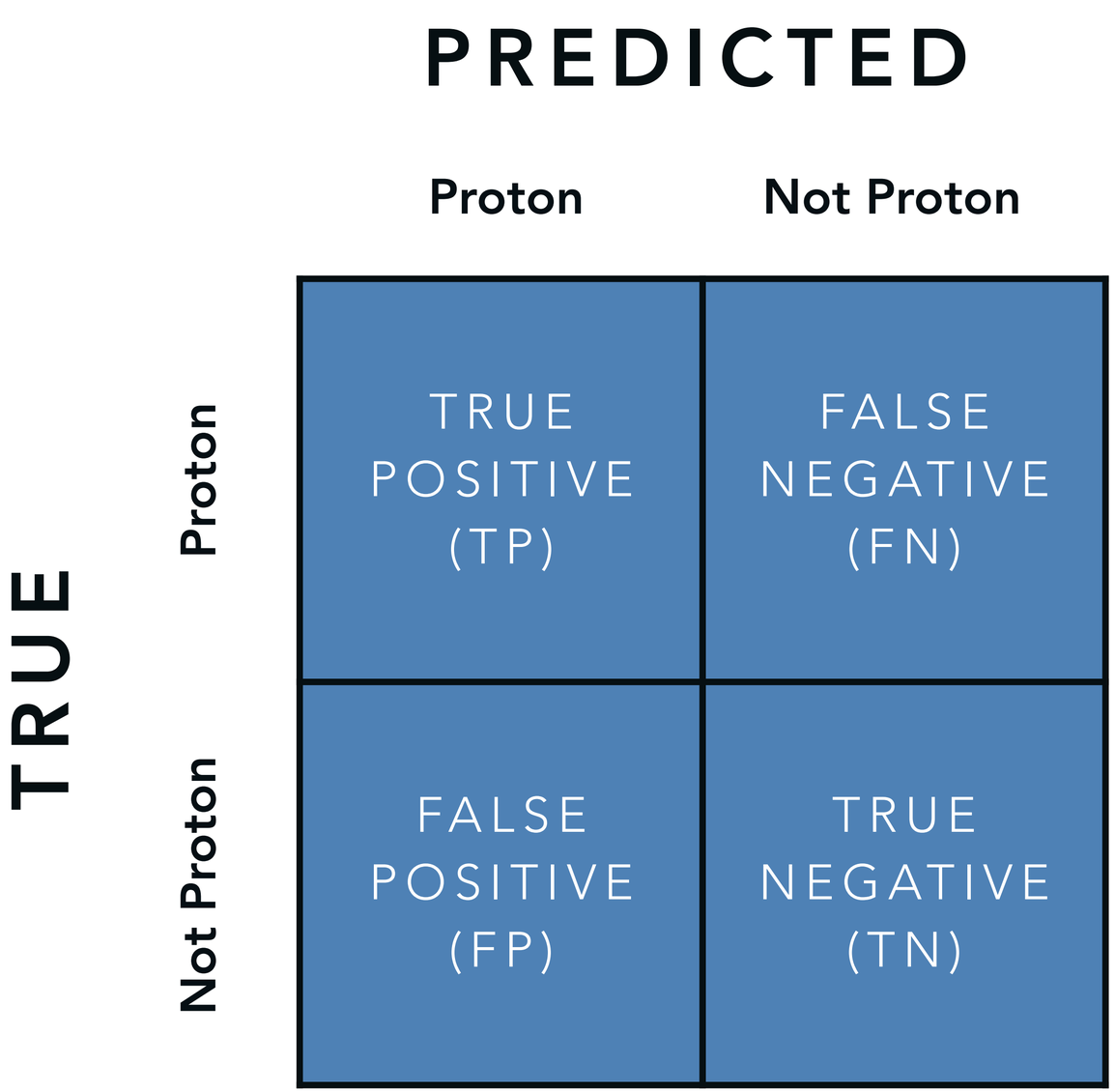
$$F1 = \frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

PREDICTED

TRUE

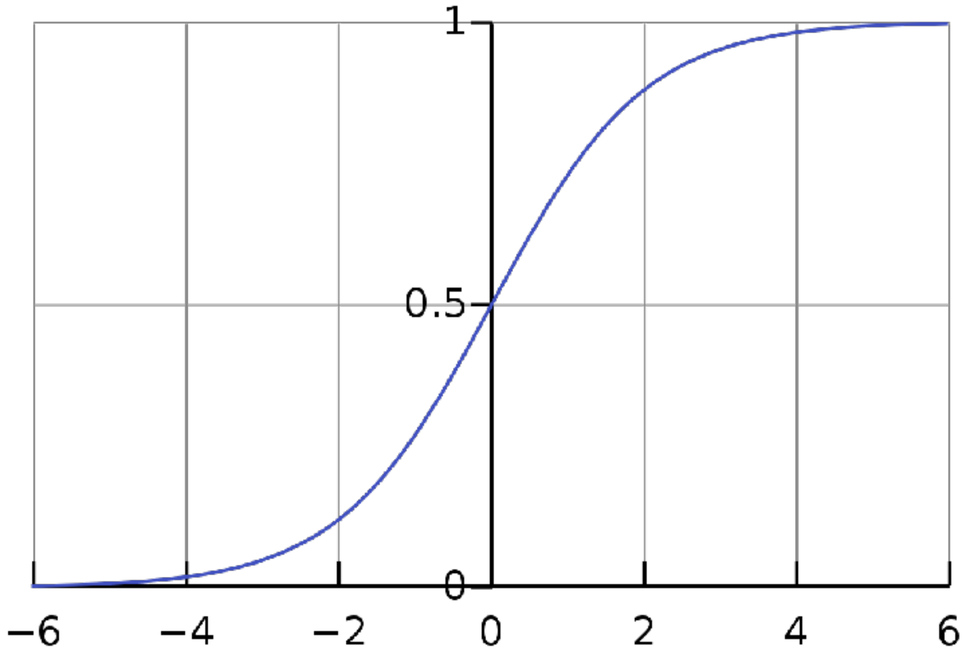
| | Proton | Not Proton |
|---|---|---|
| Proton | TRUE POSITIVE (TP) | |
| Not Proton | | TRUE NEGATIVE (TN) |

PERFECT MODEL



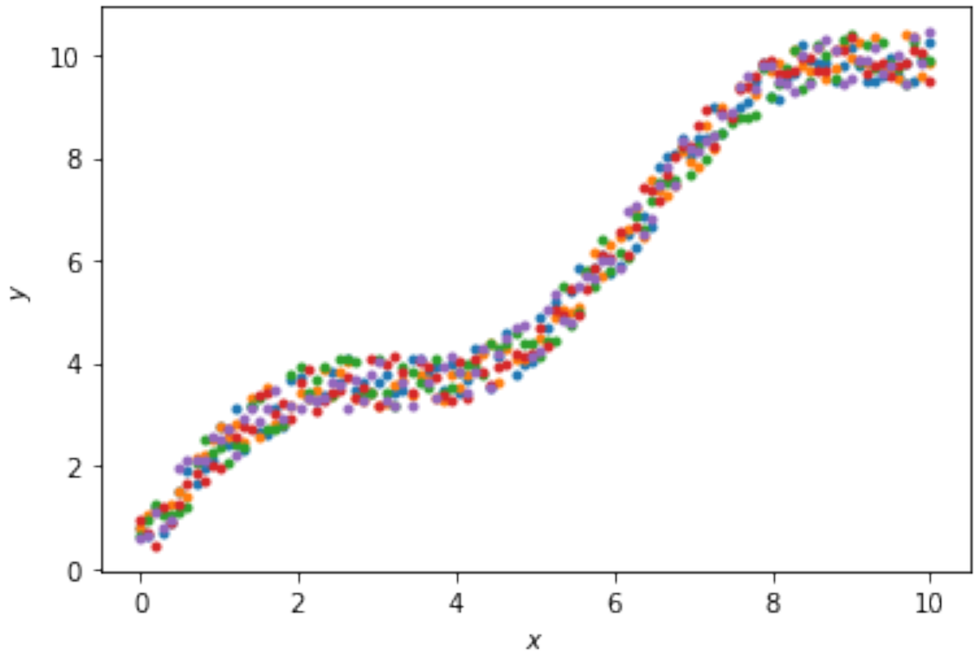

What does this matrix look like for multi-class classification?

# MULTI-CLASS CLASSIFICATION




$$\text{Softmax}(\mathbf{x})_i = \frac{e^{x_i}}{\sum_{j=1}^{C} e^{x_j}}$$

# LOSS FUNCTIONS




**Loss function**

Mean squared error (MSE)

$$J(w) = \frac{1}{N}\sum_{i=0}^{N}(\hat{y}_i - y_i)^2$$

Mean absolute error (MAE)

$$J(w) = \frac{1}{N}\sum_{i=0}^{N}|\hat{y}_i - y_i|$$

Cross entropy (CE)

$$\mathscr{L}(\mathbf{y}, \hat{\mathbf{y}}) = -\sum_{i=1}^{N}\sum_{j=1}^{C} y_{ij} \cdot \log(\hat{y}_{ij})$$

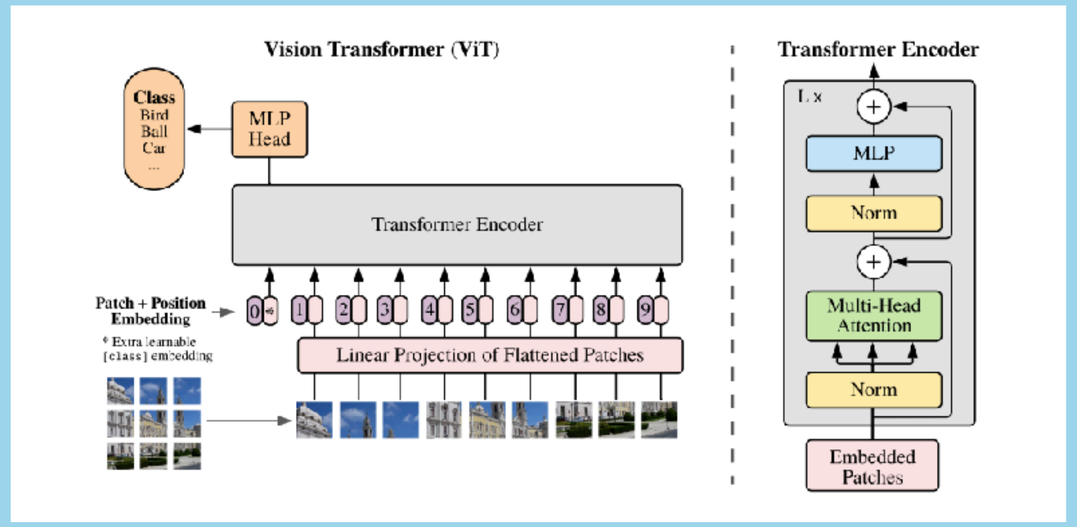
## Another approach:

# Vision Transformers

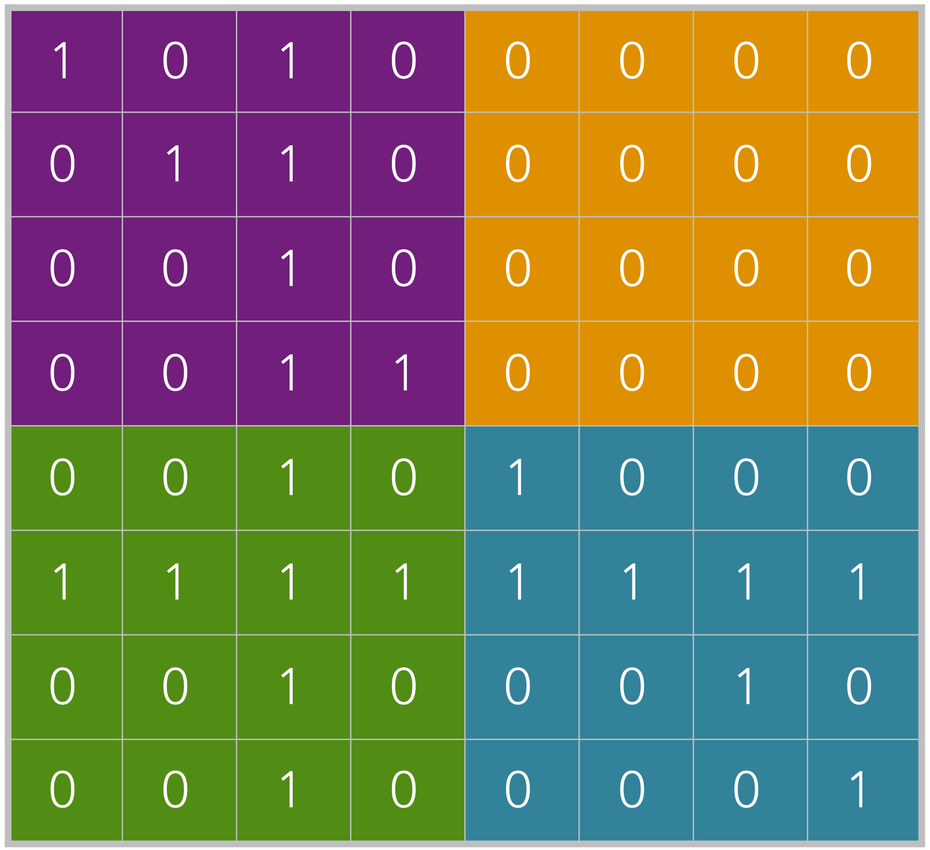
AN IMAGE IS WORTH 16X16 WORDS:
TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

**Alexey Dosovitskiy*,†, Lucas Beyer*, Alexander Kolesnikov*, Dirk Weissenborn*,
Xiaohua Zhai*, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer,
Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, Neil Houlsby*,†**

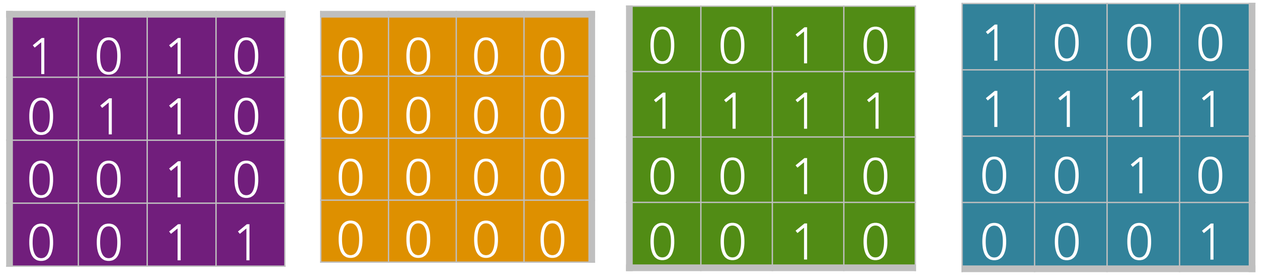
*equal technical contribution, †equal advising
Google Research, Brain Team
{adosovitskiy, neilhoulsby}@google.com



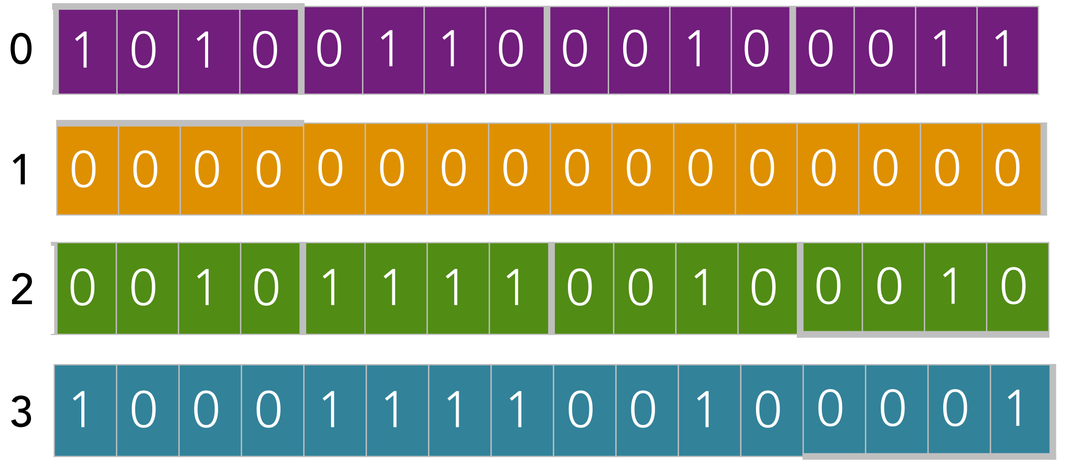
# Vision Transformer



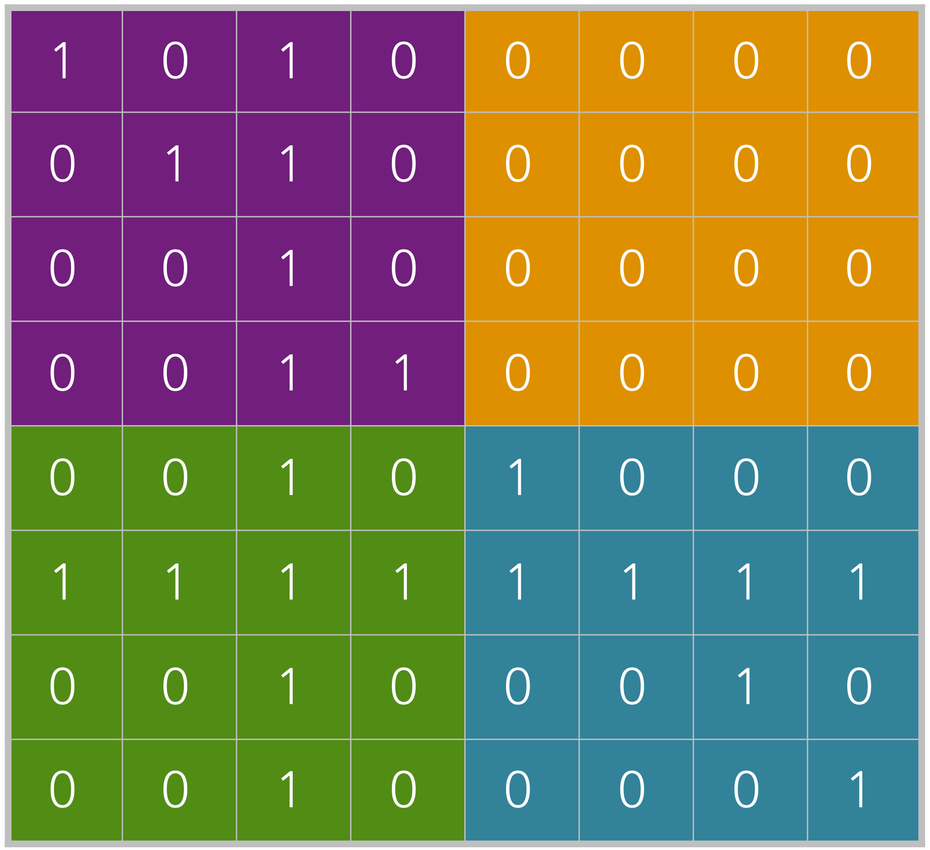

## Patches



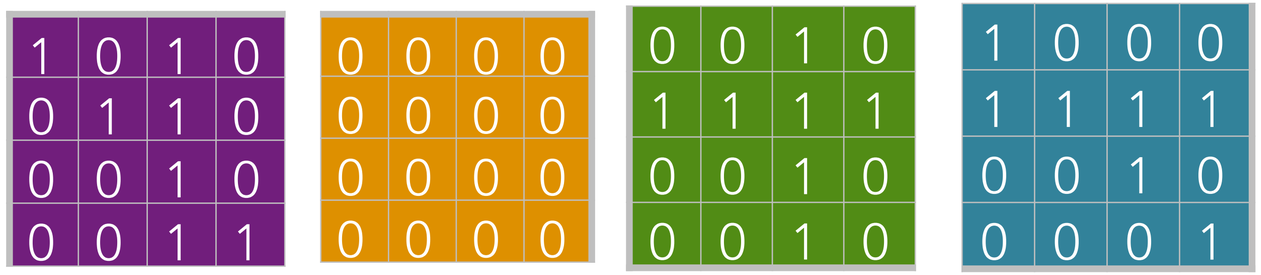

## Linear “embedding”



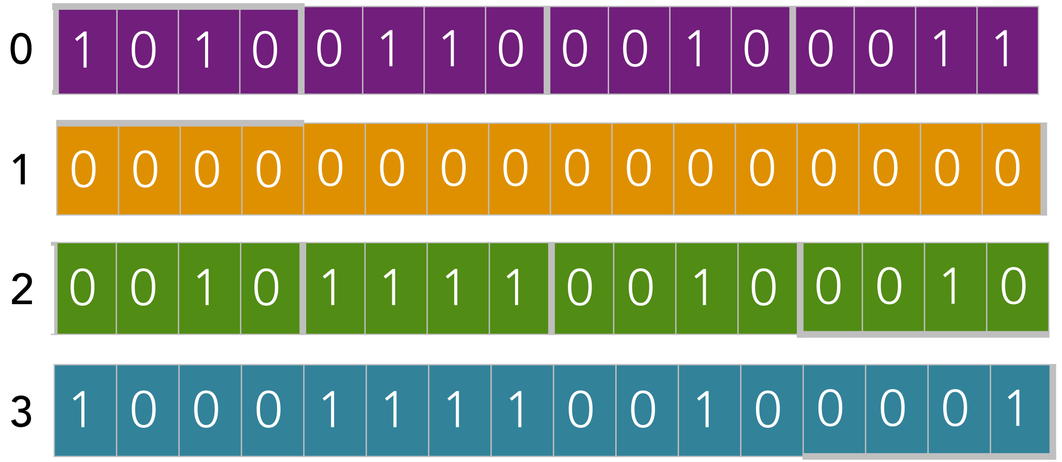
# Vision Transformer




## Patches

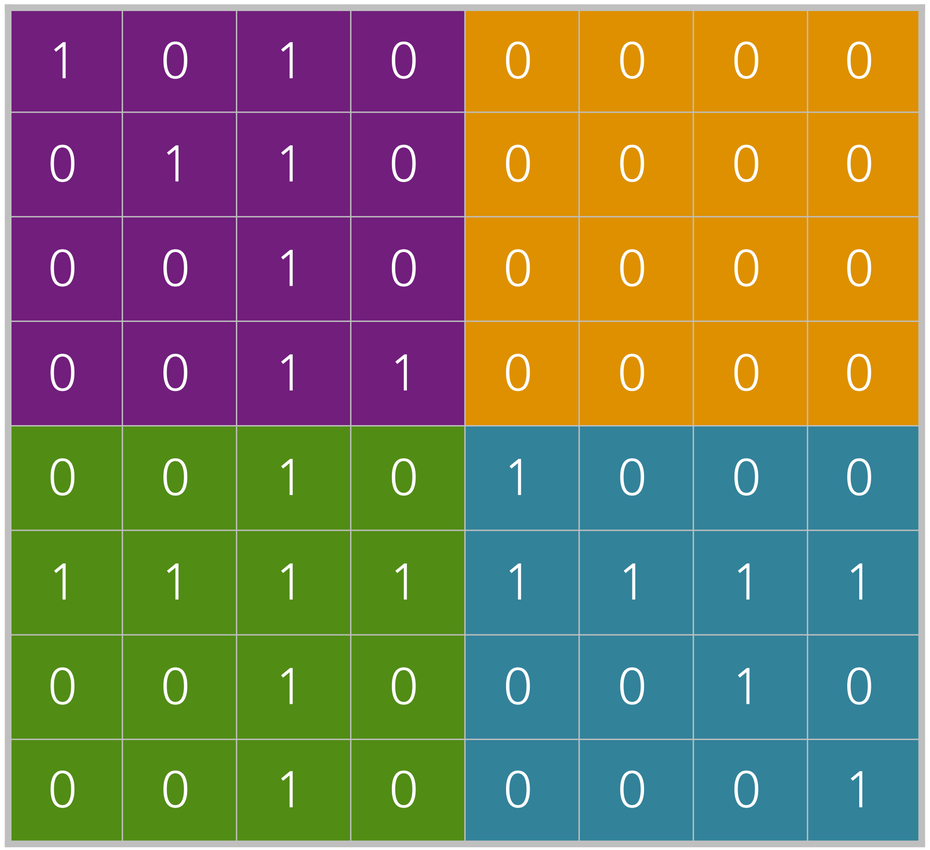



## Linear “embedding” with position encoding

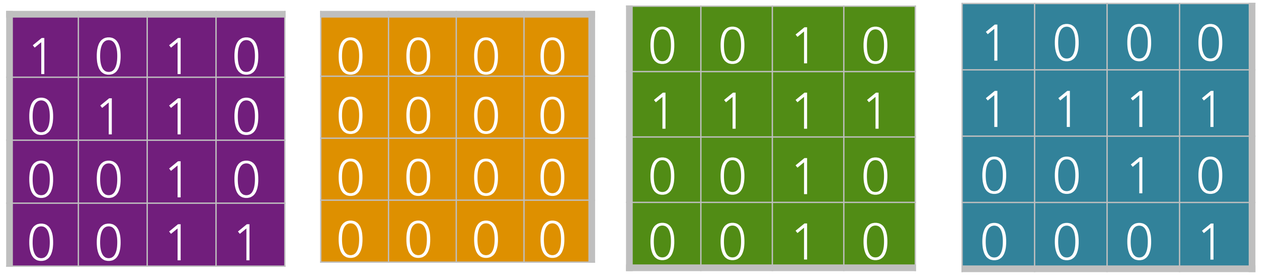



$$X = X_{patch} + P$$

# Vision Transformer

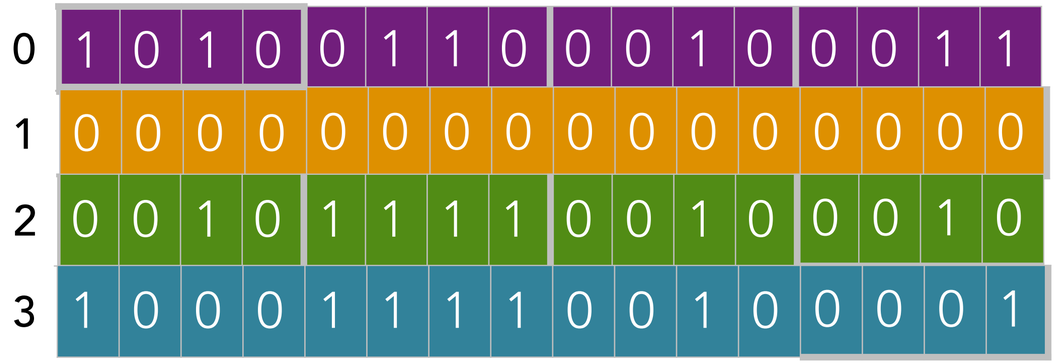



## Patches




## Linear embedding with position

$$X =$$

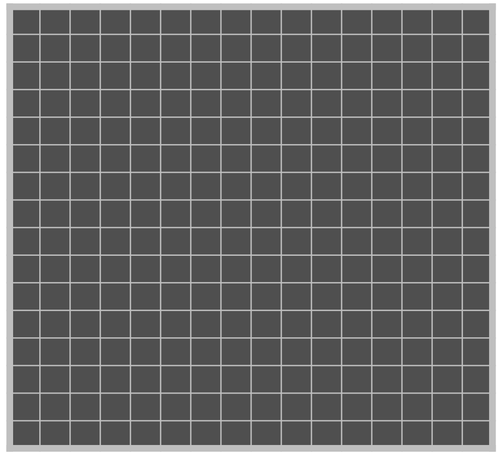


# Vision Transformer

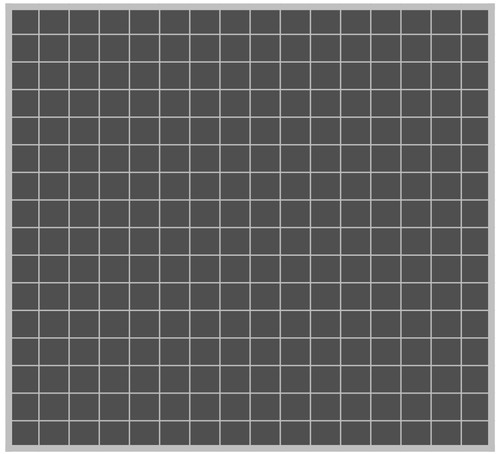
$X =$

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| 3 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |

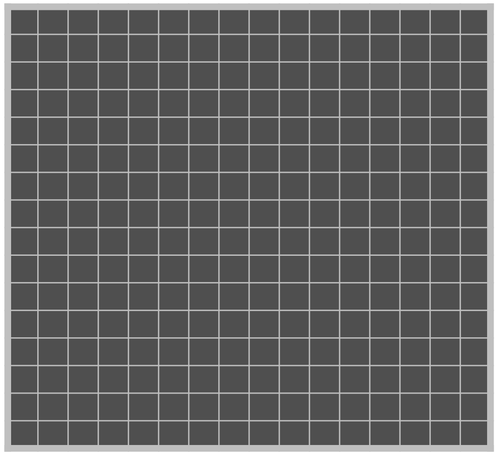
$N \times E$

$W_Q =$ 

$E \times E$

$W_K =$ 

$E \times E$

$W_V =$ 

$E \times E$

# Vision Transformer

$$X =$$

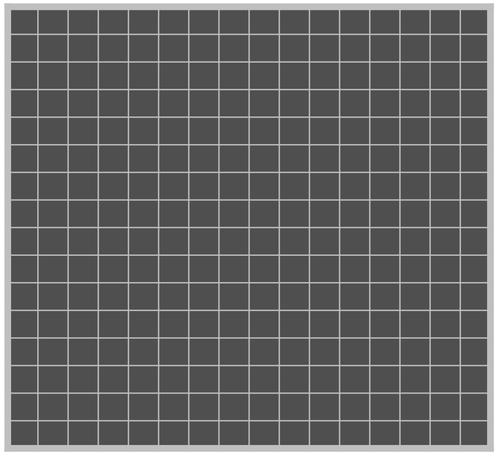
| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| 3 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |

$$Q = XW_Q$$

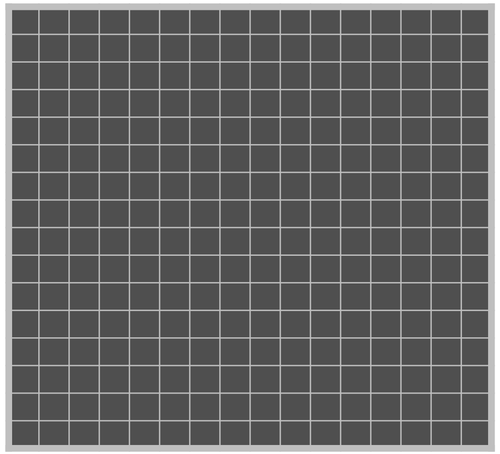
$$K = XW_K$$

$$V = XW_V$$

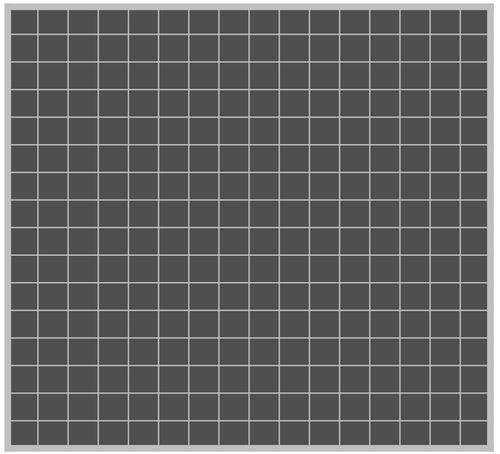
$$W_Q =$$



$$W_K =$$



$$W_V =$$



# Vision Transformer

$$X =$$

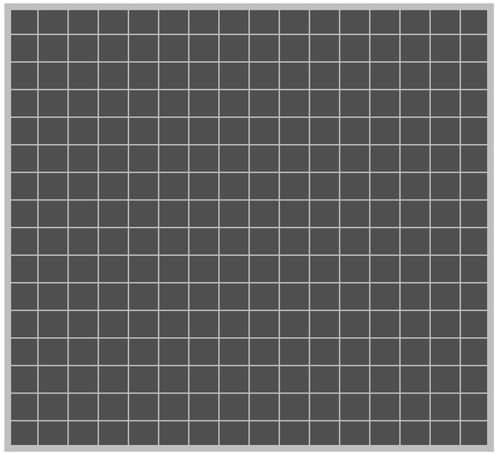
| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| 3 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |

$$Q = XW_Q \qquad (N \times E)$$

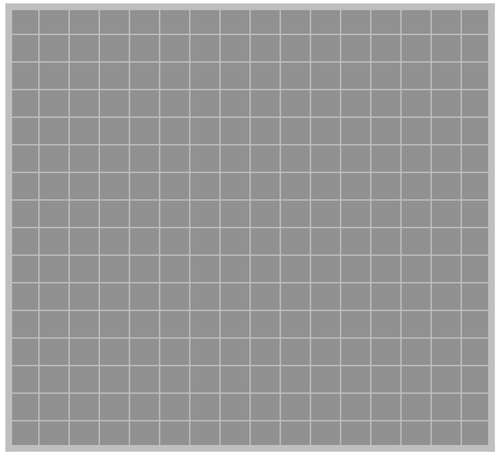
$$K = XW_K \qquad (N \times E)$$

$$V = XW_V \qquad (N \times E)$$

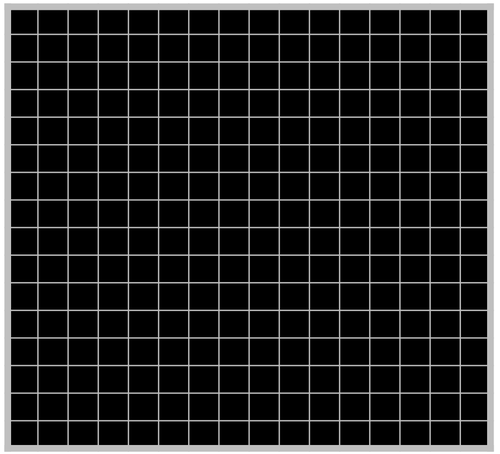
$$W_Q =$$



$$W_K =$$



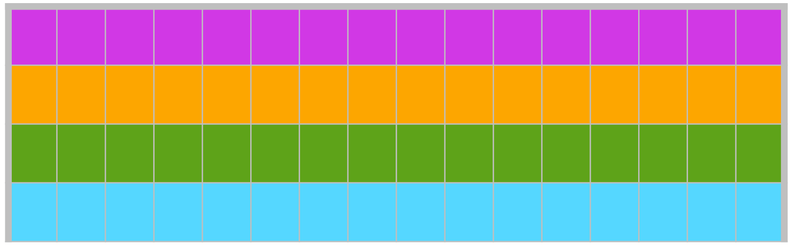
$$W_V =$$



# Vision Transformer

$$Q = XW_Q =$$

$$K = XW_K =$$

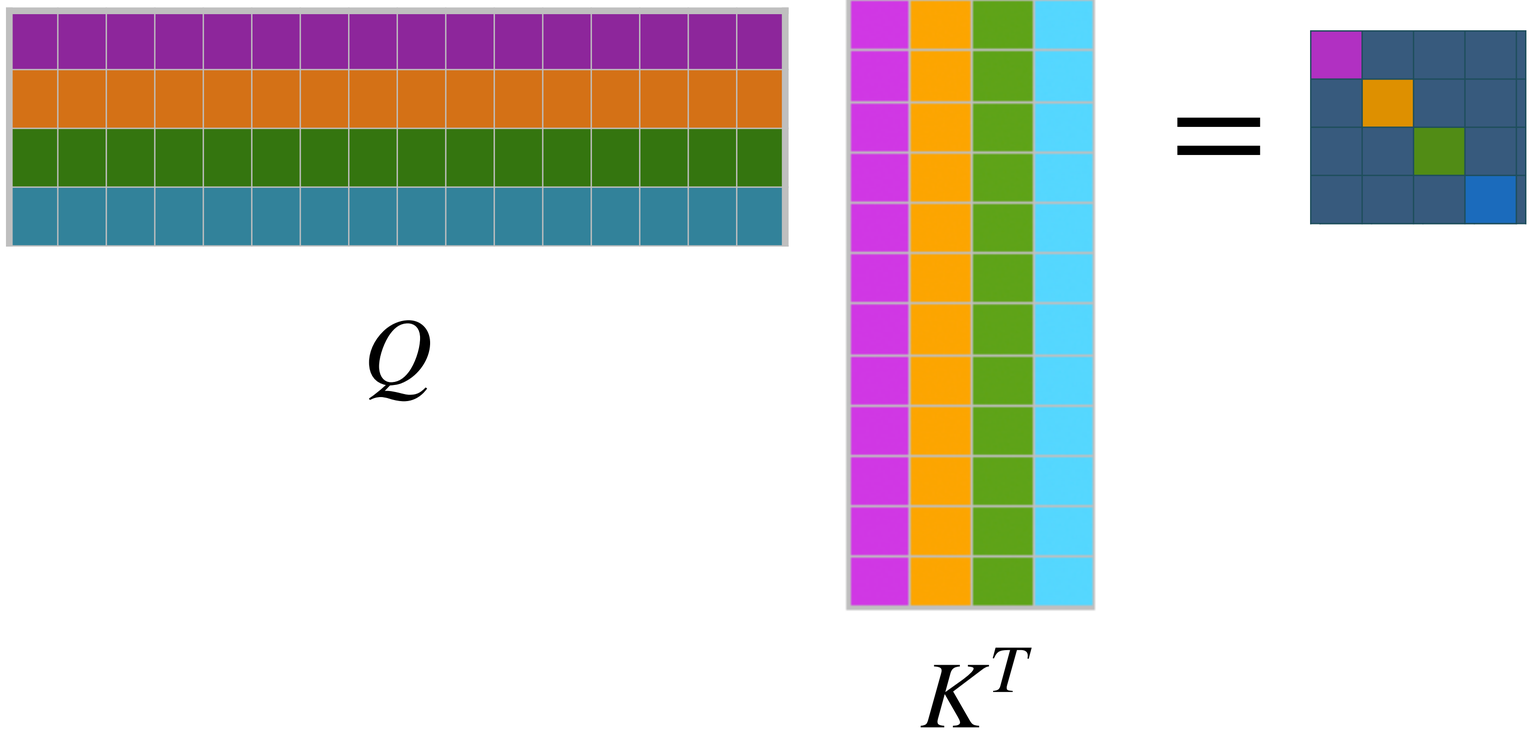


$$V = XW_V =$$

$$\text{scores} = QK^T$$

# Vision Transformer

$$\text{scores} = QK^T$$




## Attention:

$$\text{A} = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})$$

## Output

$$Z = AV$$

# Vision Transformer

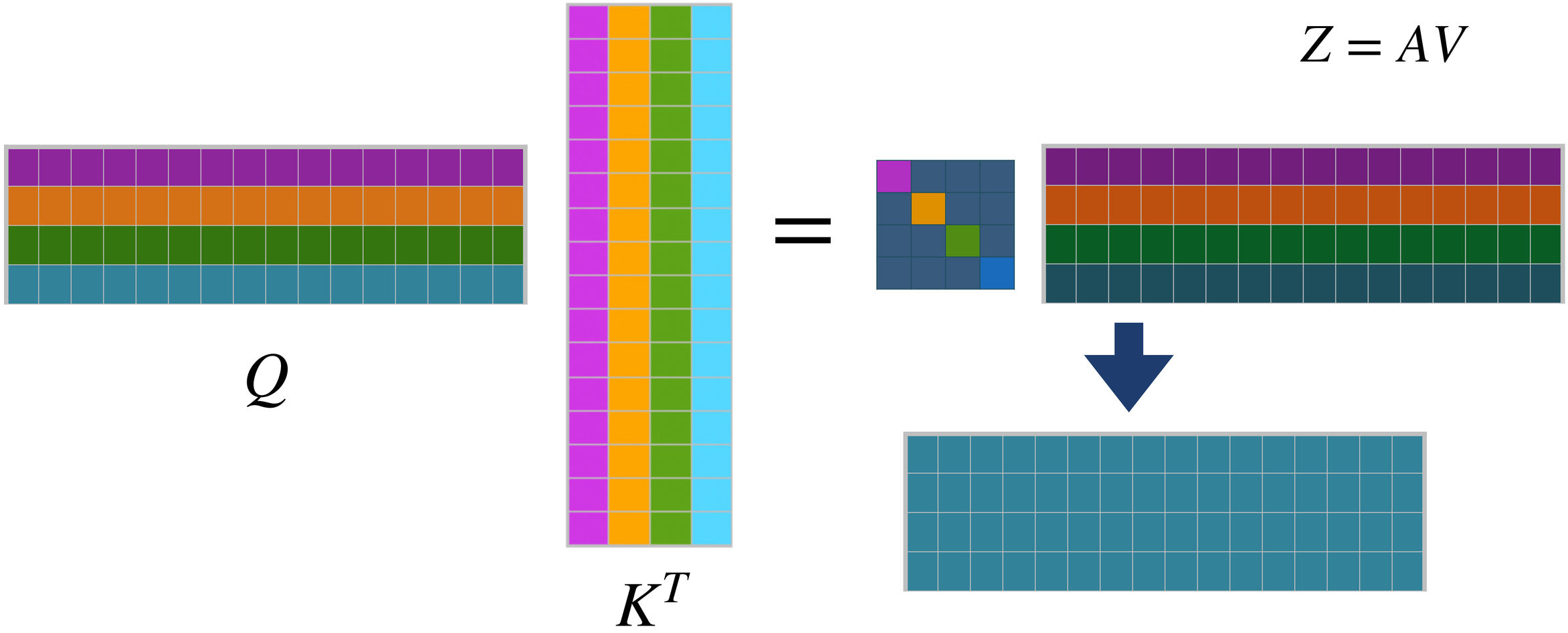
$$\text{scores} = QK^T$$

**Attention:**

$$\text{A} = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})$$

**Output**

$$Z = AV$$

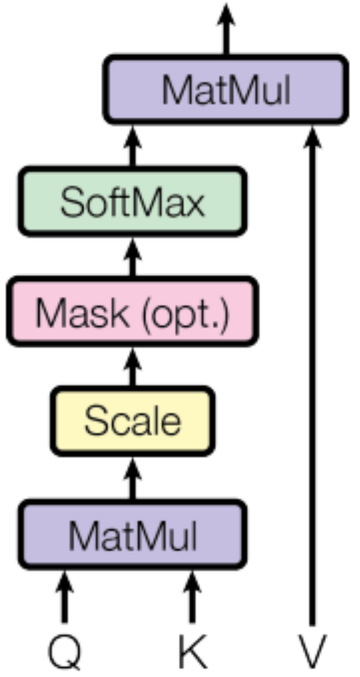


# Vision Transformer

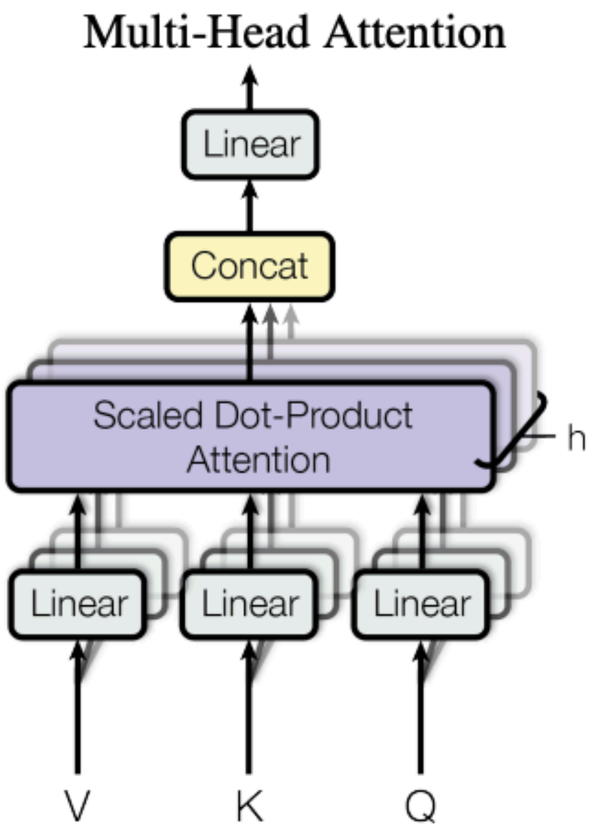
## Output

$$Z = AV$$

Scaled Dot-Product Attention



# Vision Transformer




**Attention Is All You Need**

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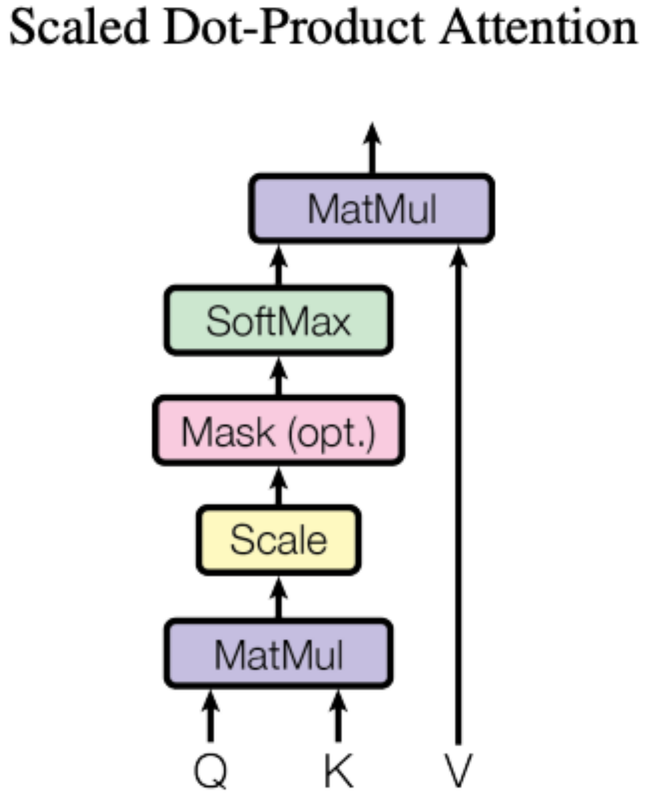


# CHOOSING AN ARCHITECTURE

HOW MANY LAYERS?

HOW MANY NODES PER LAYER?

LEARNING RATE

DROPOUT?

WHAT ACTIVATION FUNCTION(S)?

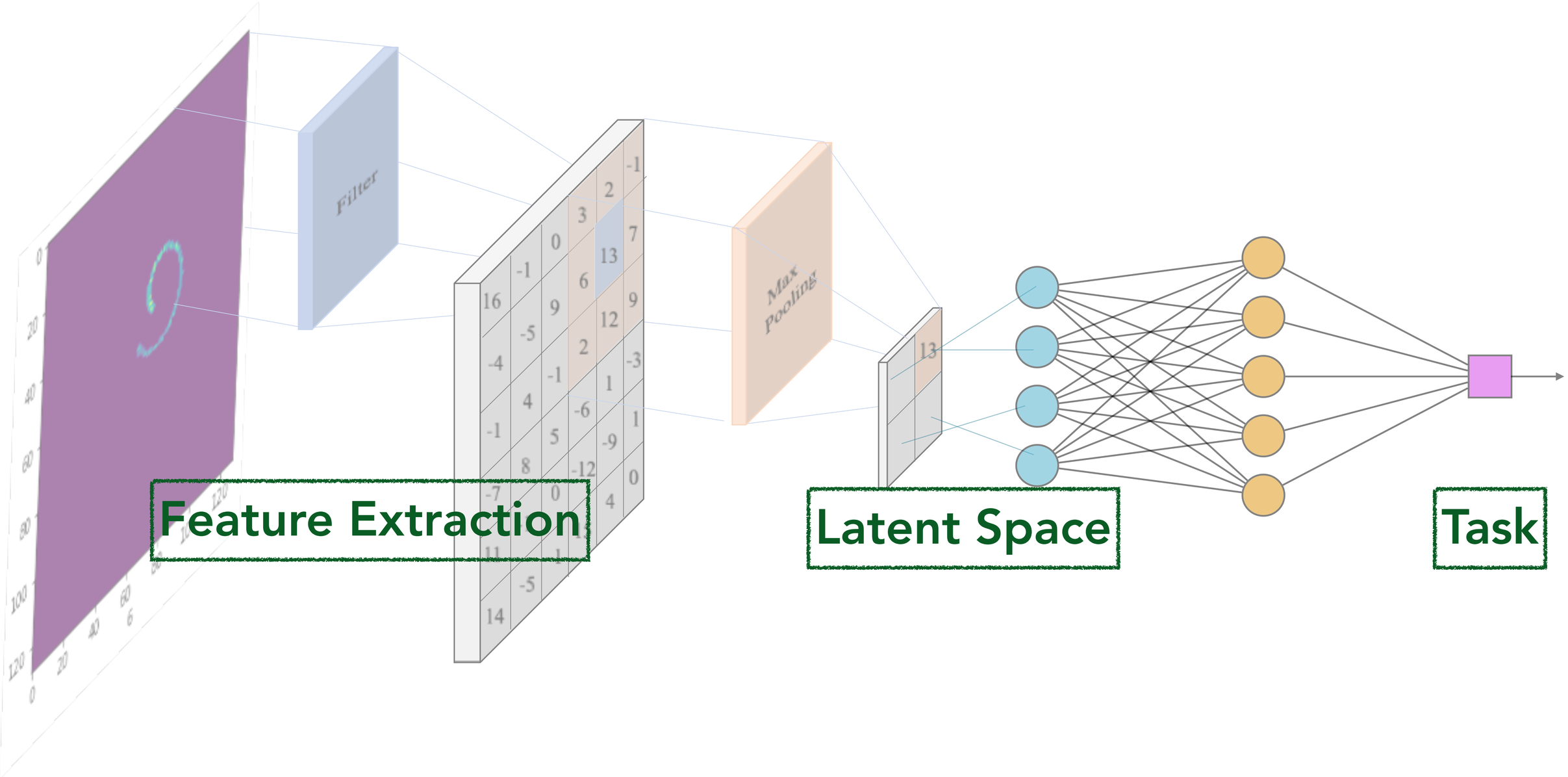
HOW MANY CONVOLUTION LAYERS?

FILTER SIZE?

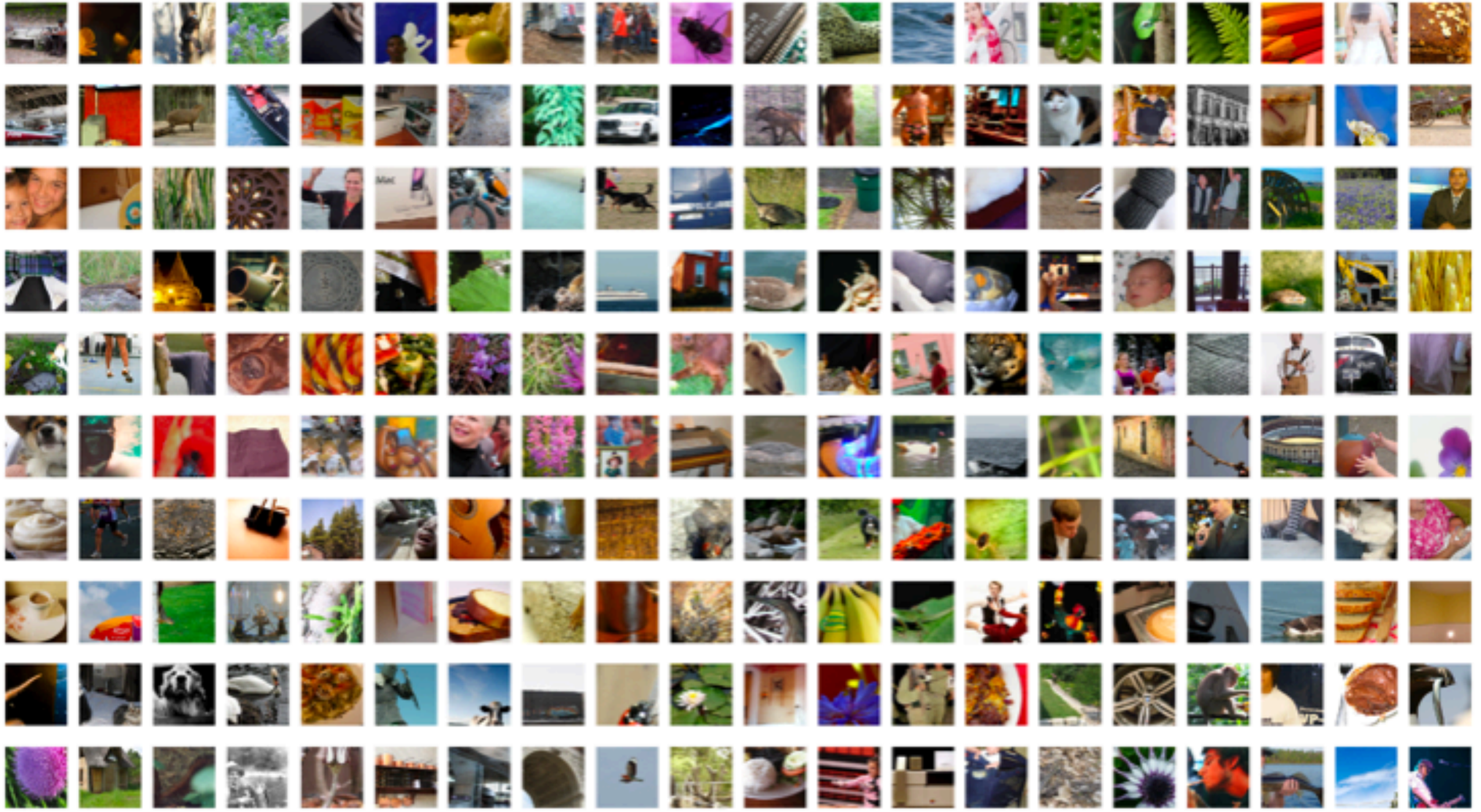
STRIDE?

POOLING?

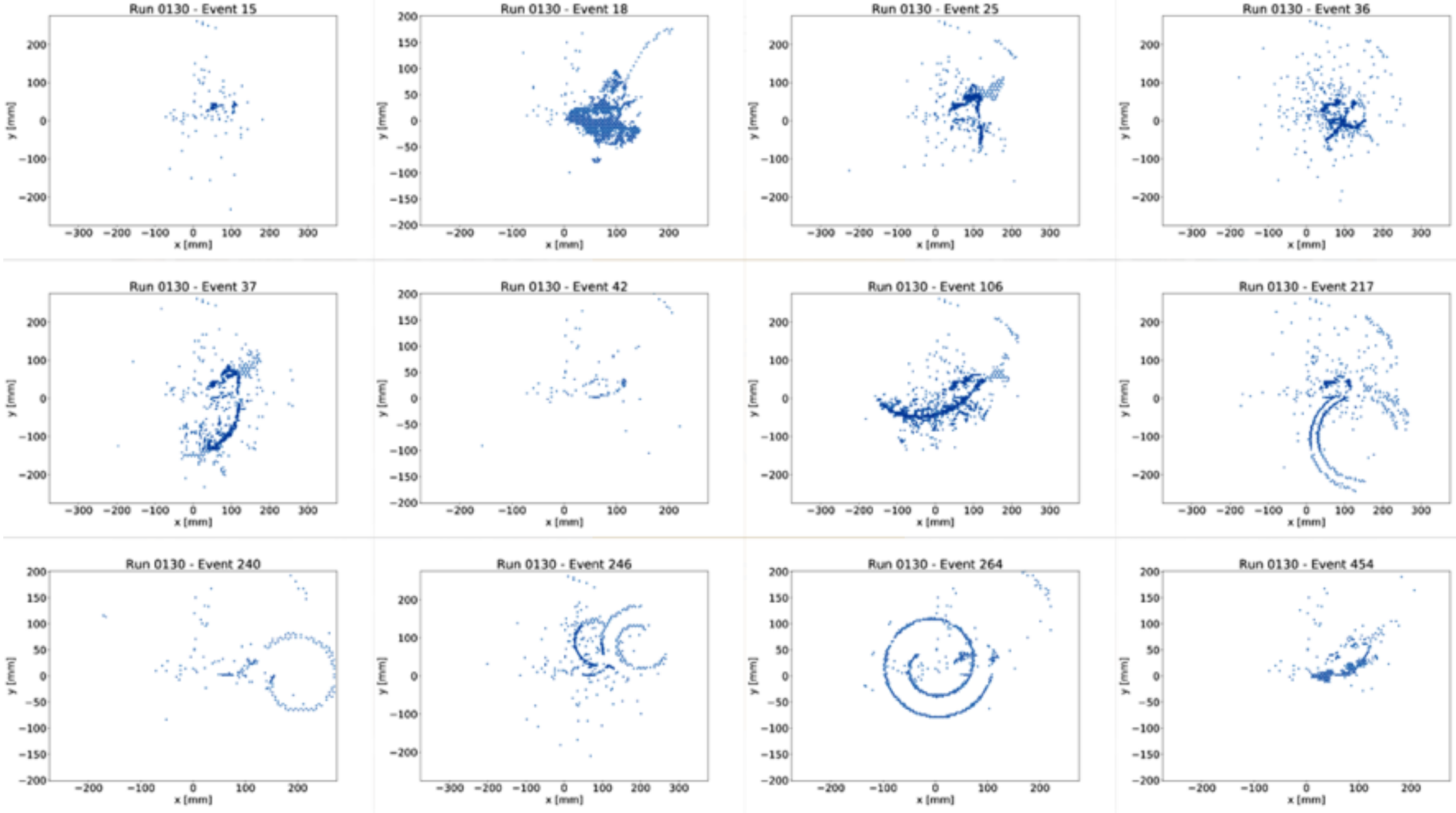
# PRE-TRAINED MODELS

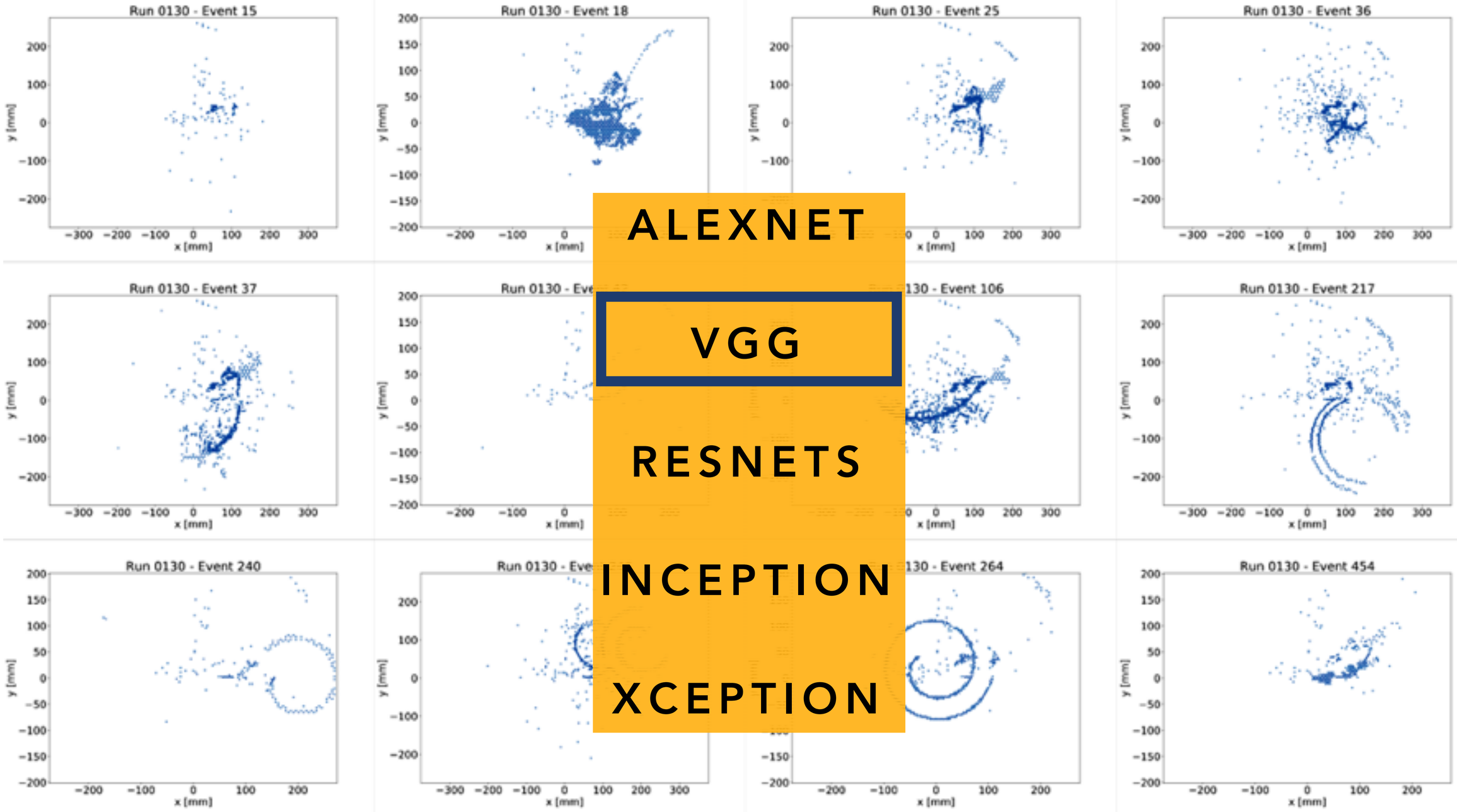


# PRE-TRAINED MODELS



# PRETRAINED MODELS


Run 0130 - Event 15
Run 0130 - Event 18
Run 0130 - Event 25
Run 0130 - Event 36
Run 0130 - Event 37
Run 0130 - Event 42
Run 0130 - Event 106
Run 0130 - Event 217
Run 0130 - Event 240
Run 0130 - Event 246
Run 0130 - Event 264
Run 0130 - Event 454
x [mm]
y [mm]




ALEXNET

VGG

RESNETS

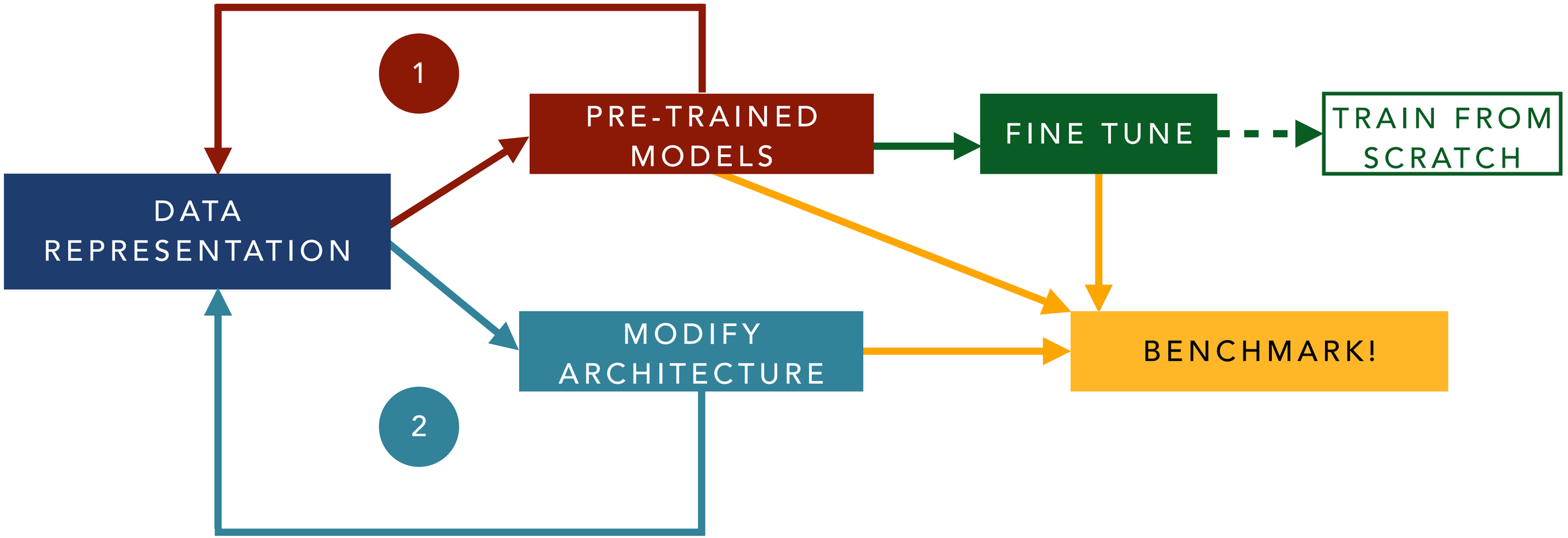
INCEPTION

XCEPTION

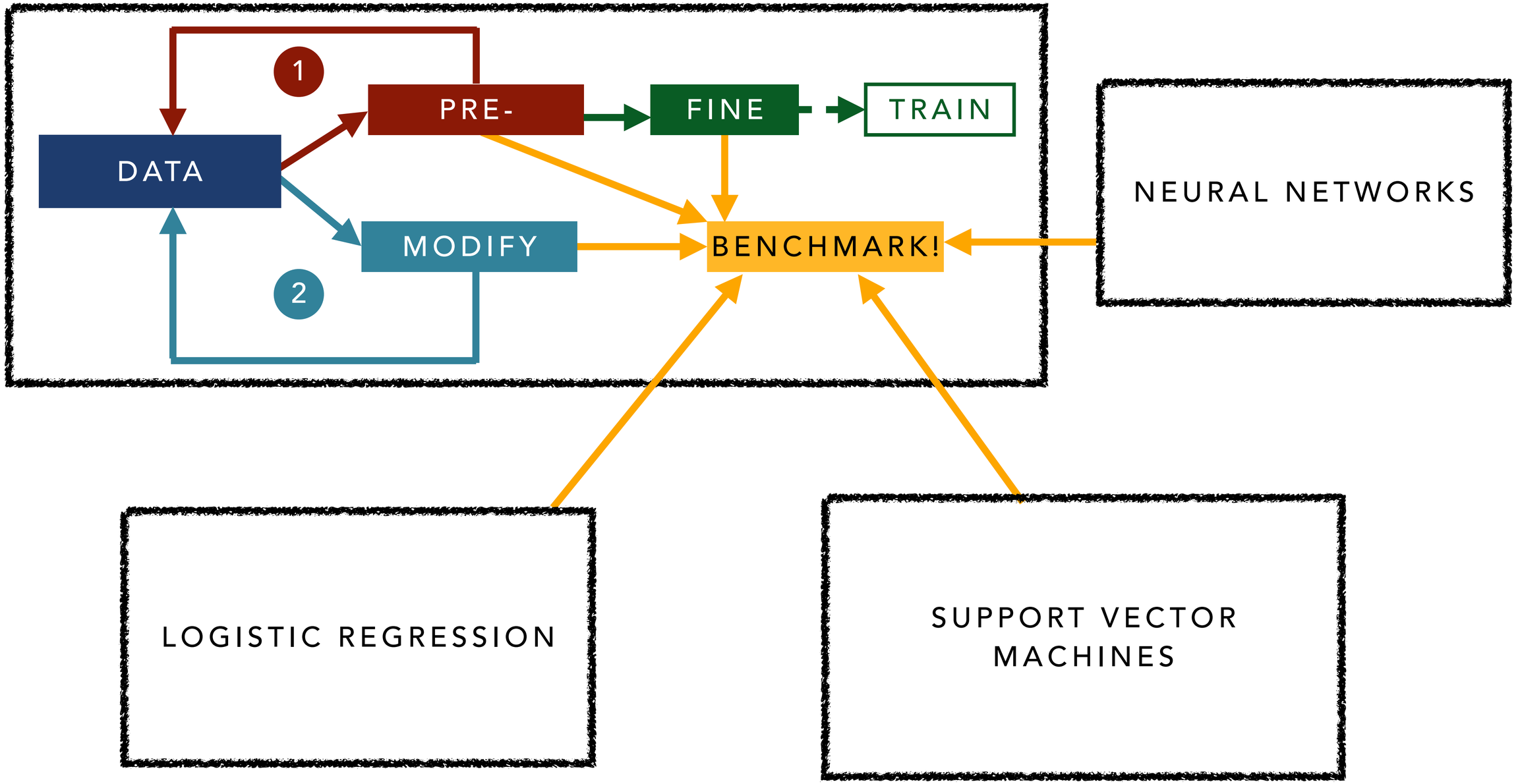
J. Z. TAYLOR, HONOR'S THESIS, DAVIDSON COLLEGE

M.P. KUCHERA, ET. AL. NUCLEAR INSTRUMENTATION AND METHODS A (2019)

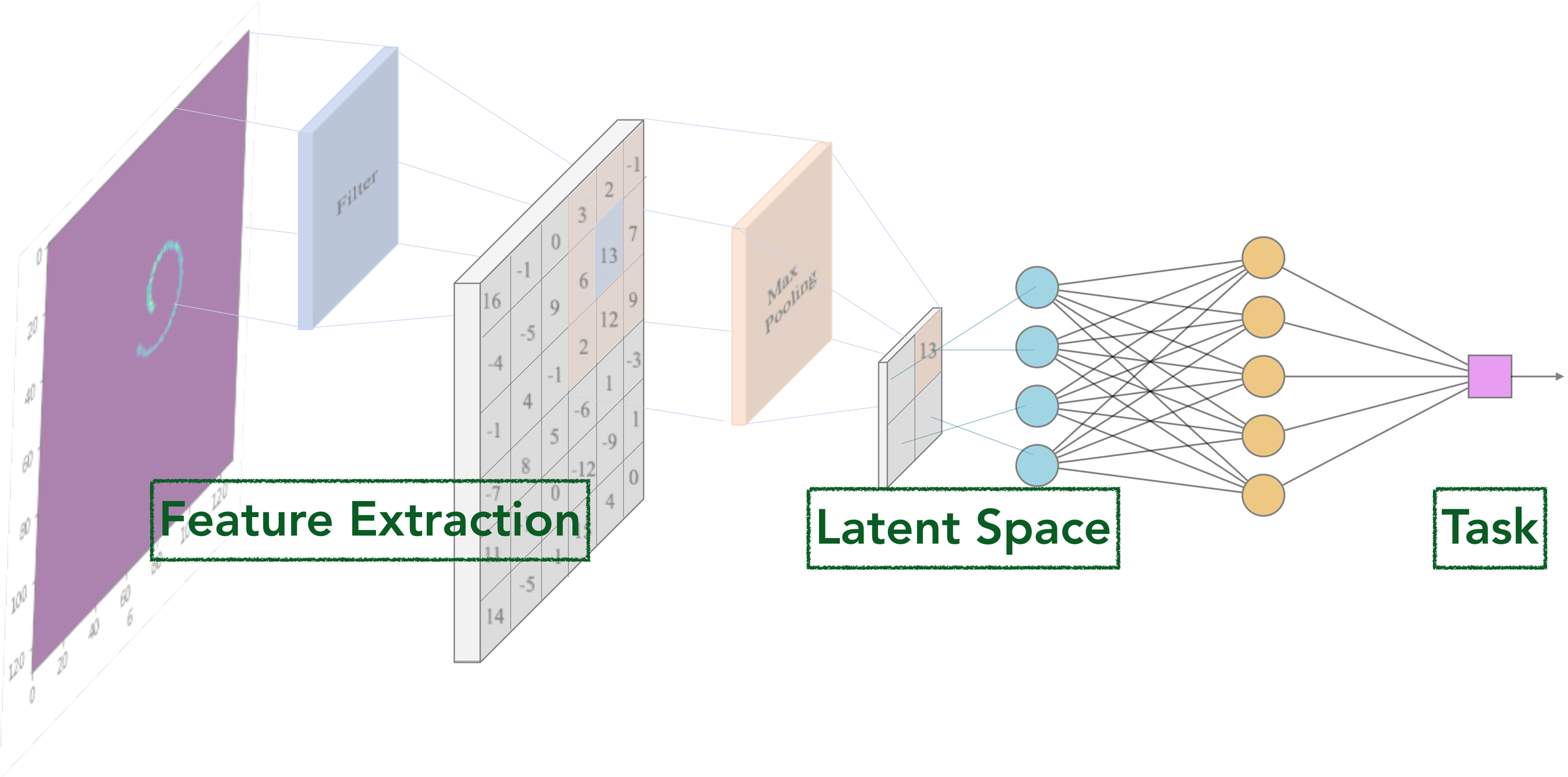
# EXAMPLE WORKFLOW



# EXAMPLE WORKFLOW



# CONVOLUTIONAL NEURAL NETWORKS



# OTHER NETWORK MECHANISMS:

## -POINT CONVOLUTIONS

## -SYMMETRIES